Hysteresis Indicies and Beta of IBW sites: Curry, Silverado, and Lake Marguerite

2026-03-26

Aims to plot hysteresis per analyte per storm per site

Metrics

- *Hysteresis index* is calculated as the median (+/- 95% CI) across 2% intervals of discharge.

  +HI = clockwise (source limitation or nearstream source)

  -HI = counterclockwise (transport limitation or distal source)

- *Beta* is the slope of the CQ relationship on the rising limb of the storm

  CIs on beta are bootstrapped 95% CI

NOTES

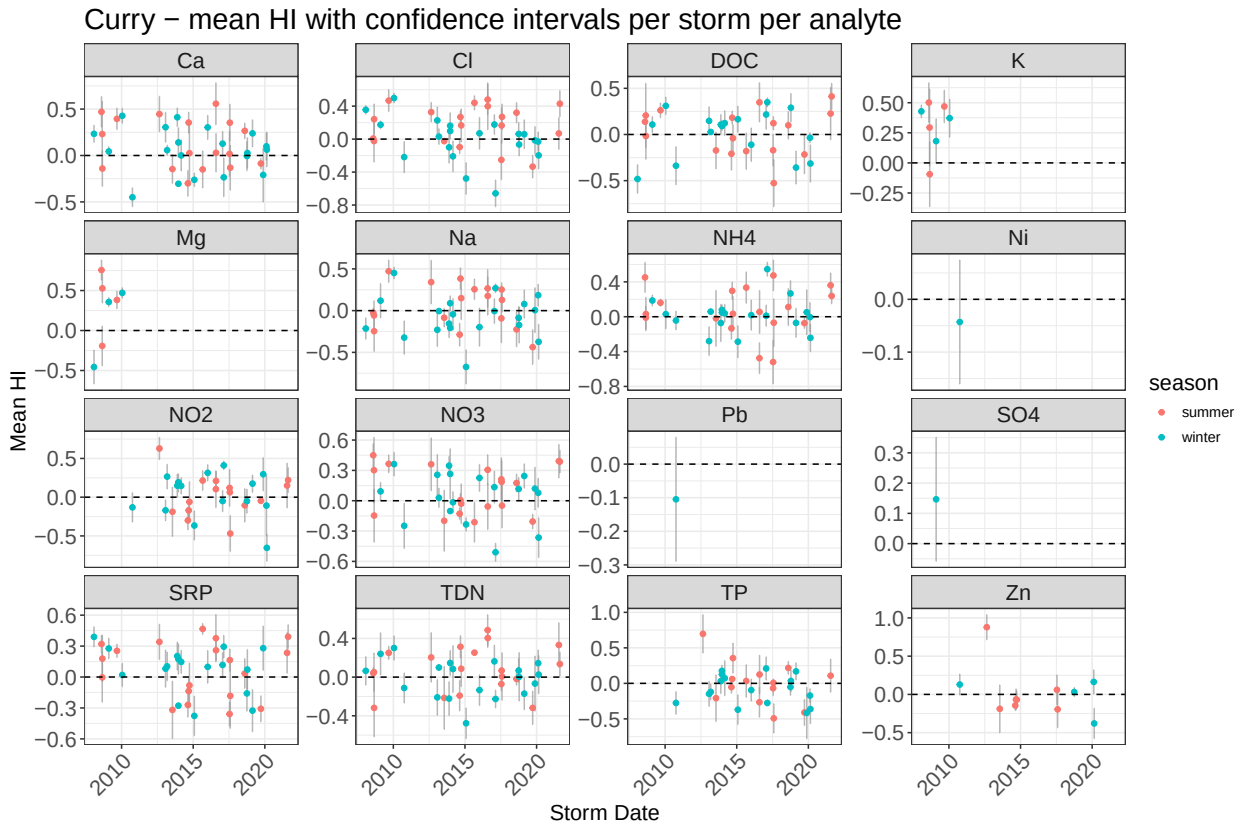
- Lake Marguerite does not have a lot of chems to work with. Do we know if there are more chems than the 624_runoff_chems.csv from the dropbox?
- Similarly, there is a gap of data in Silverado 2014-2018, was there less sampling in that time or is the data elsewhere?
- individual analyte outliers > 3sd above mean concentration per storm were removed for hysteresis and beta calculations, but remain for individual storm plotting at the bottom of the report

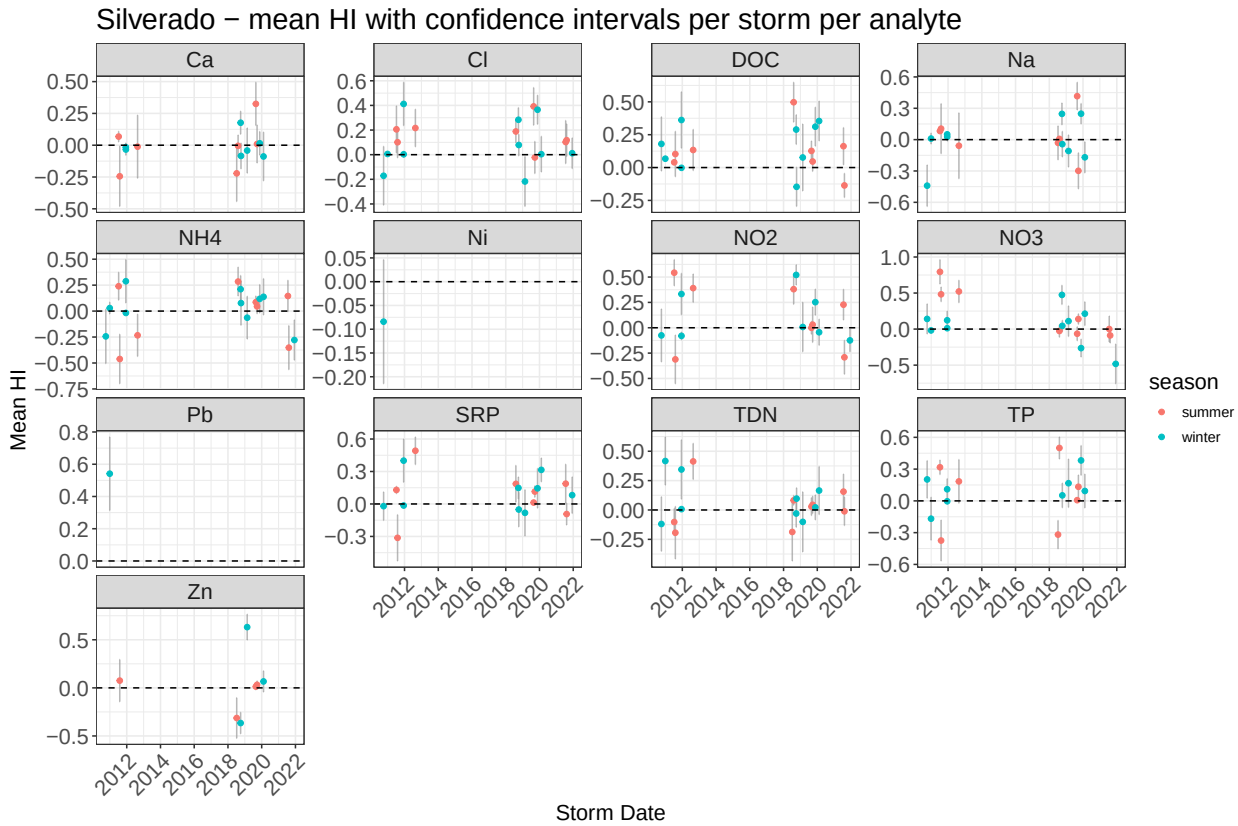
Data munging/cleaning

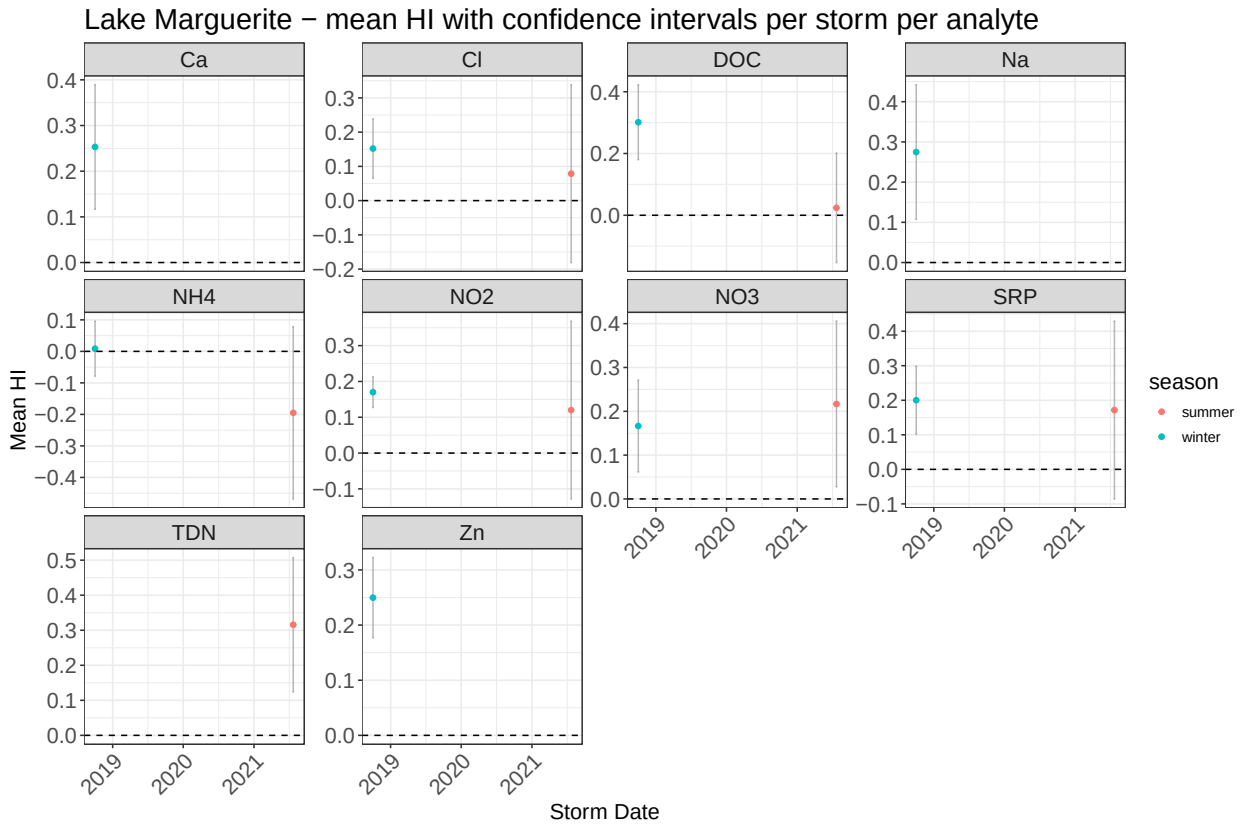
Workflow before this rmd, in order:

- q_storms.R: import discharge from Lake Margherite, Silverado, & Curry, delineate unique storms, and interpolate Q
- merge_Q_chems.R: merge discharge data with chem data per site, produce brief cq plots per analyte per site
- hysteresis_calc_2.R: calculate hysteresis based on python function, removed a lot of storms with not enough data

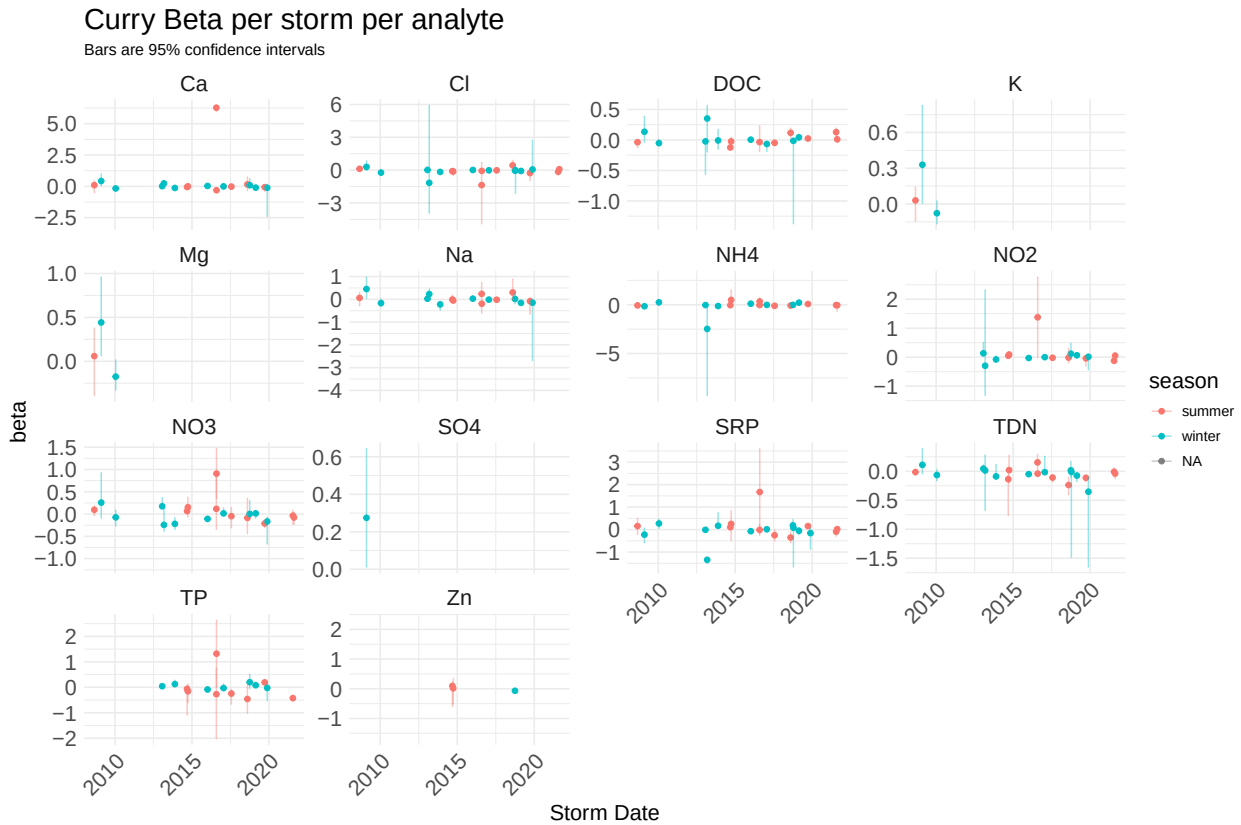
Plotting Hysteresis

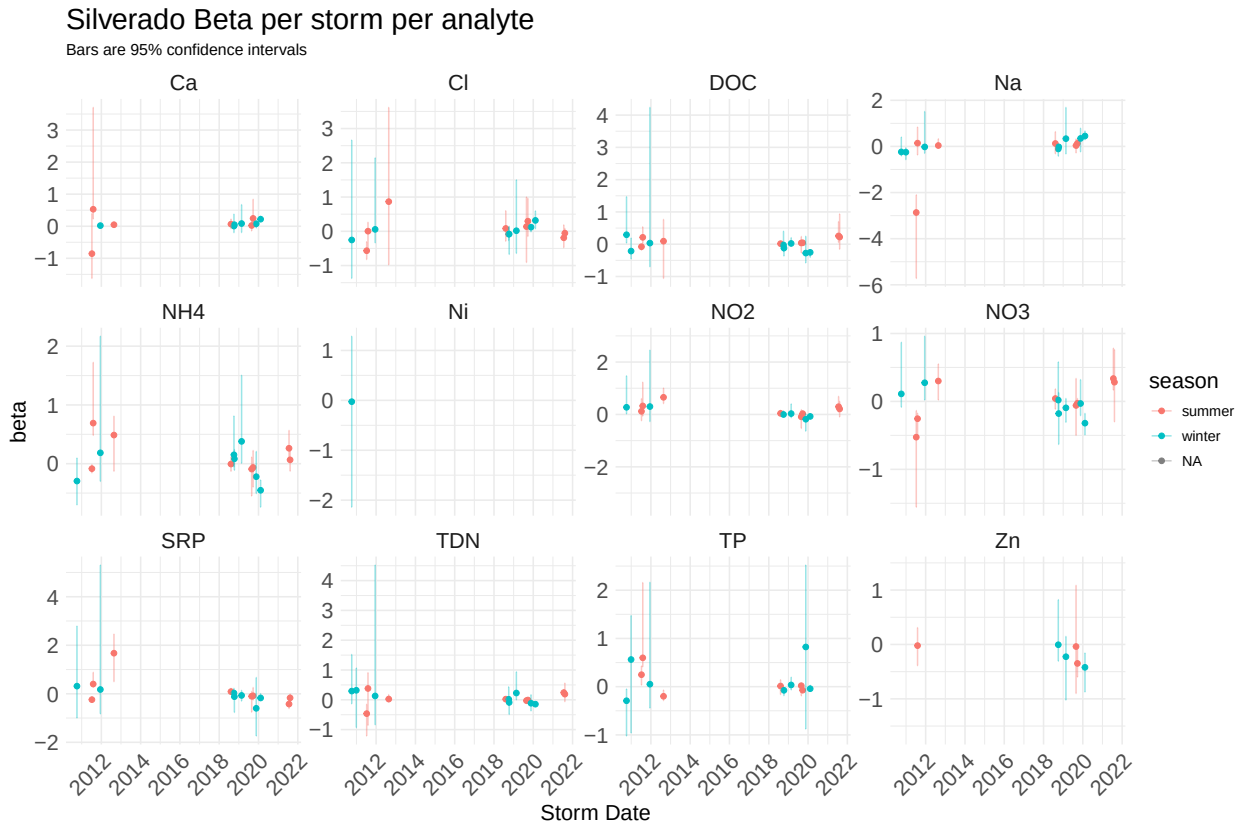


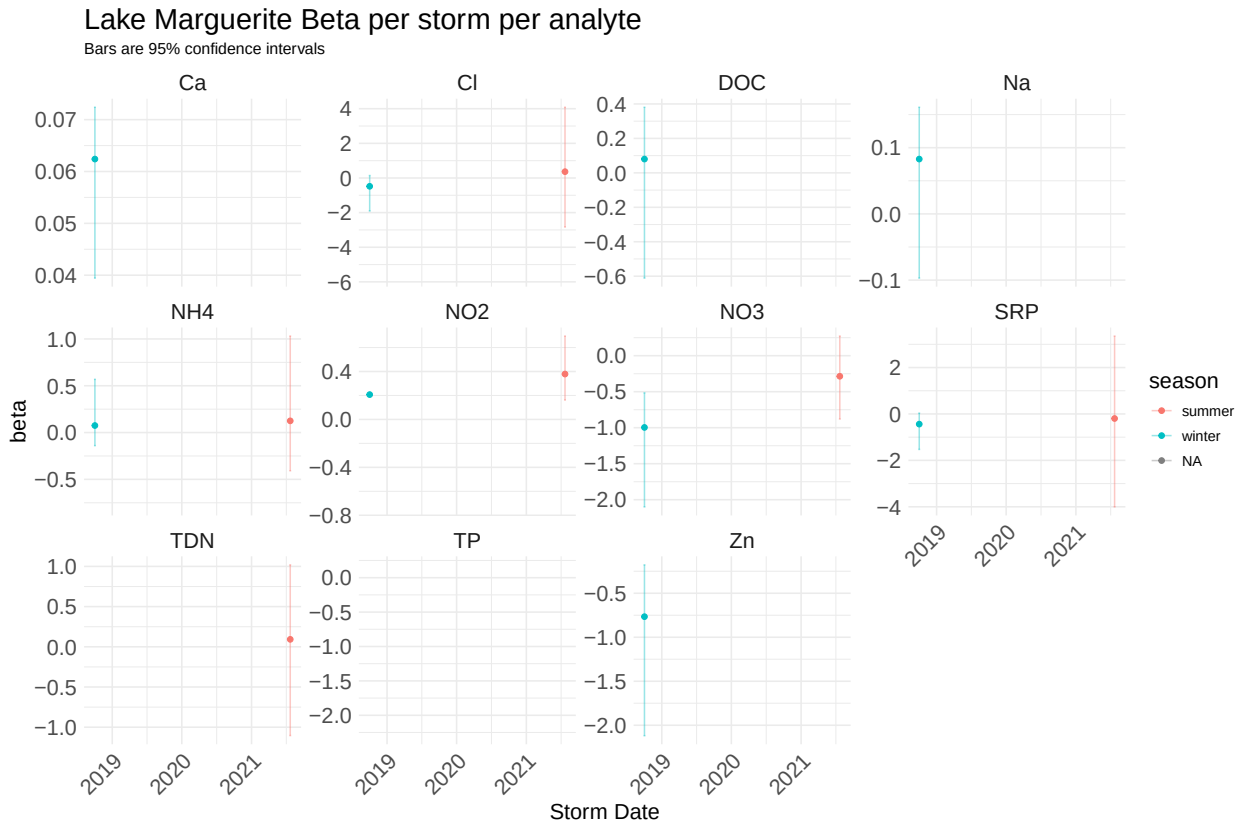


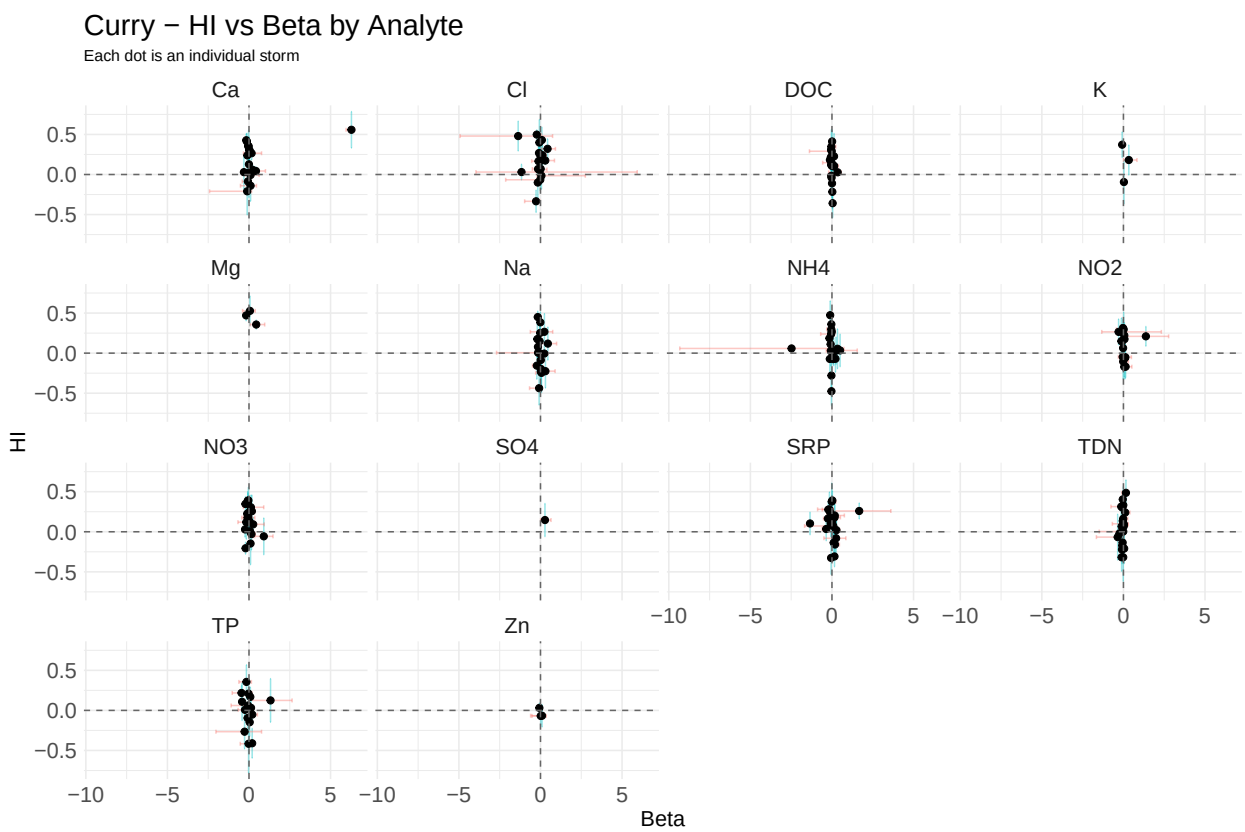

Lake Marguerite − mean HI with confidence intervals per storm per analyte
Ca
Cl
DOC
Na
NH4
NO2
NO3
SRP
TDN
Zn
Mean HI
Storm Date
2019
2020
2021
season
summer
winter

Plotting Beta

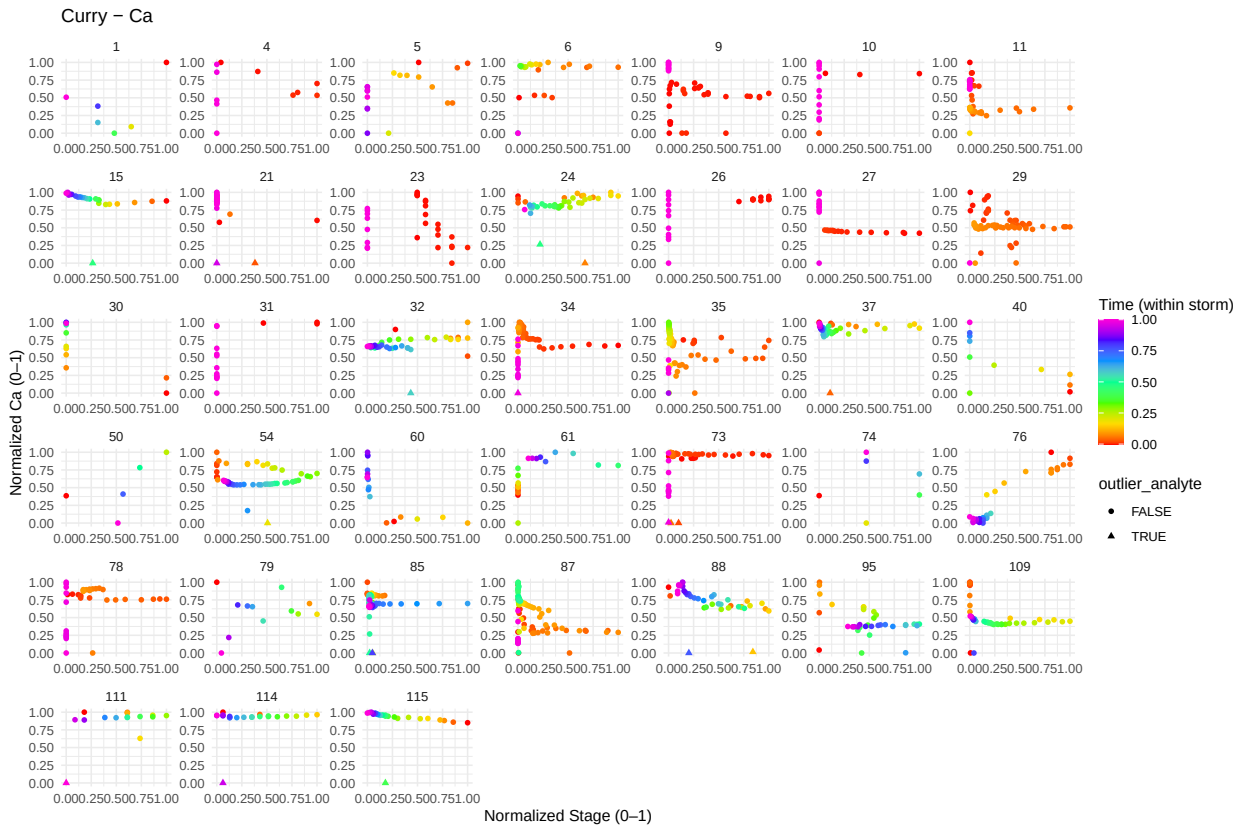


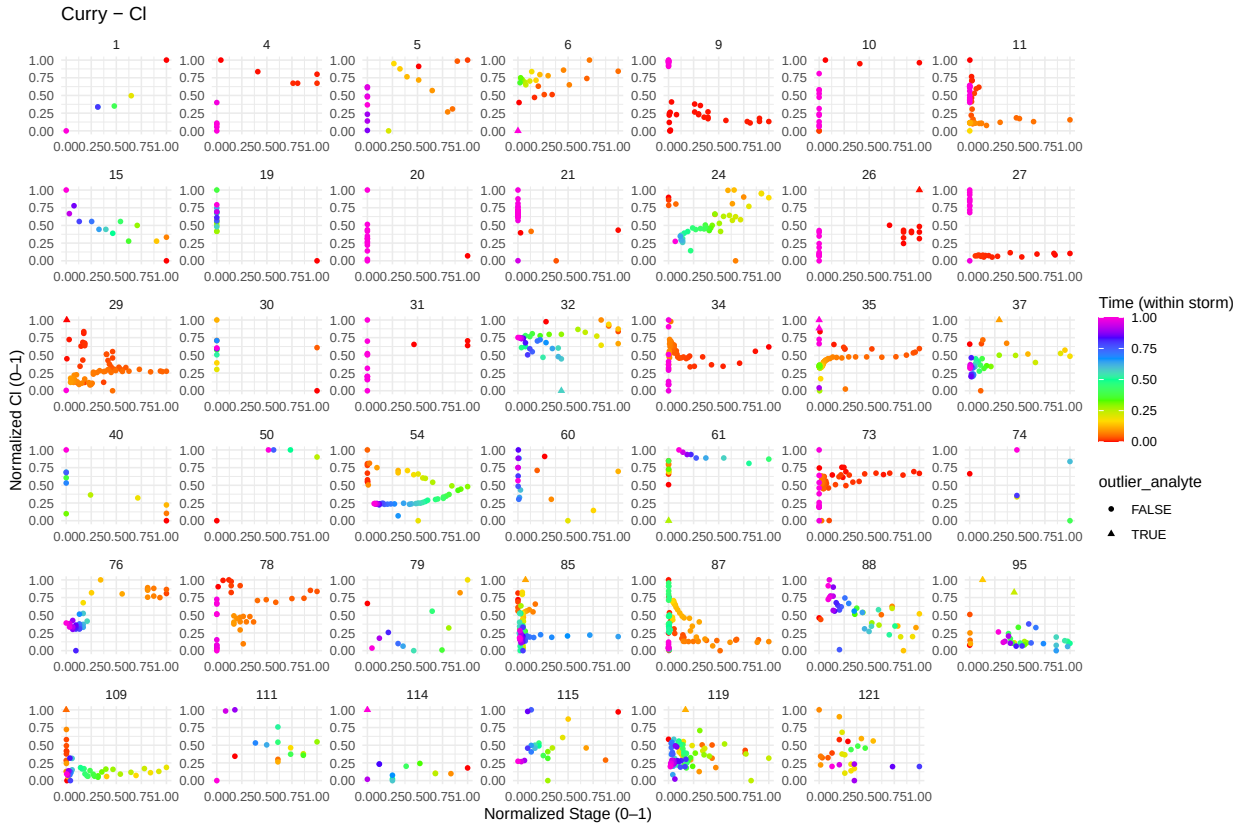



Lake Marguerite Beta per storm per analyte
Bars are 95% confidence intervals
Ca
Cl
DOC
Na
NH4
NO2
NO3
SRP
TDN
TP
Zn
beta
Storm Date
season
summer
winter
NA

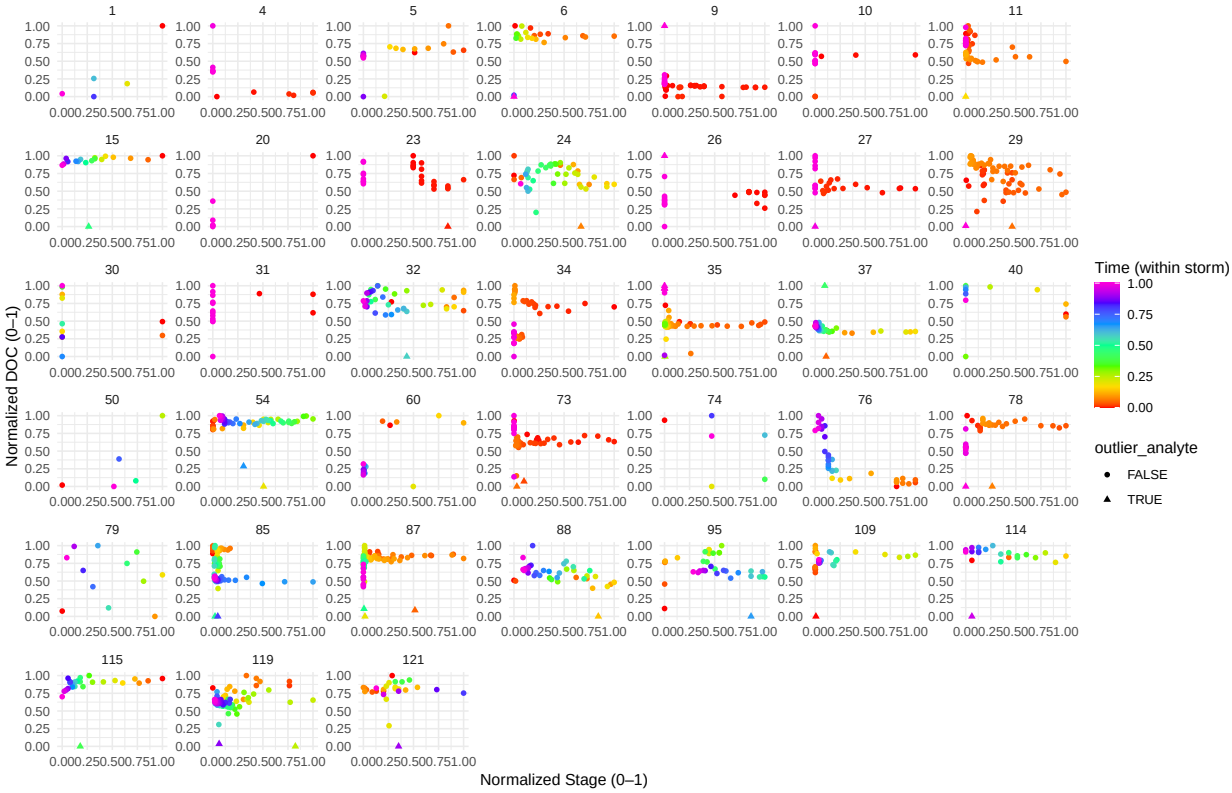

Curry − HI vs Beta by Analyte
Each dot is an individual storm
Ca
Cl
DOC
K
Mg
Na
NH4
NO2
NO3
SO4
SRP
TDN
TP
Zn
HI
Beta

Plotting CQ

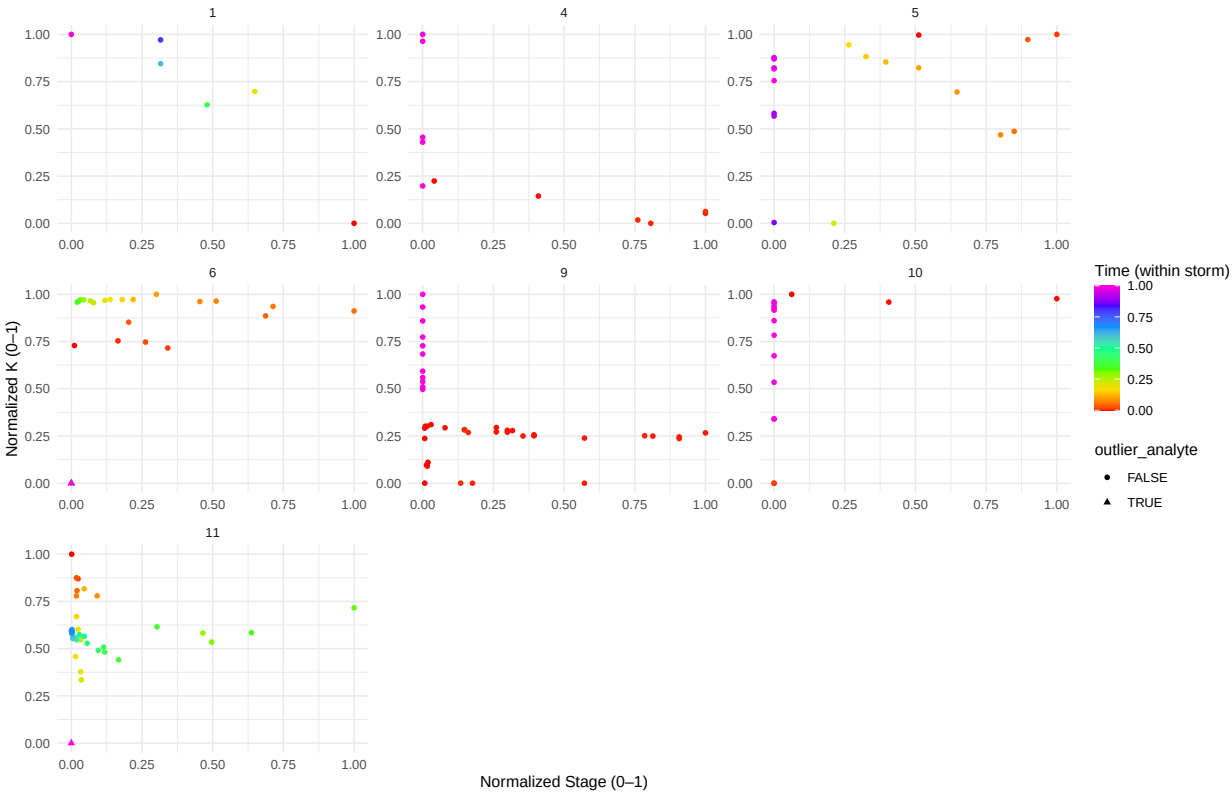

Curry − Ca
Normalized Ca (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

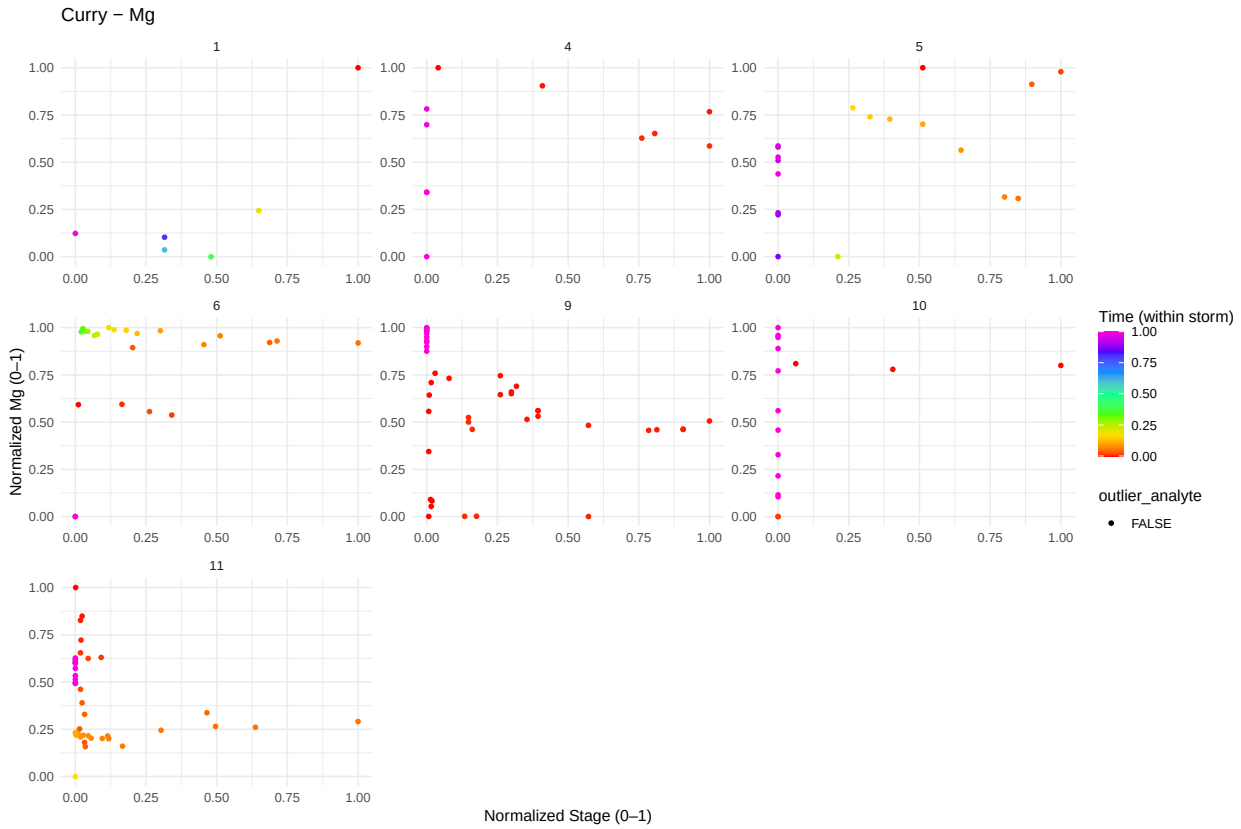

Curry − Cl
Normalized Cl (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

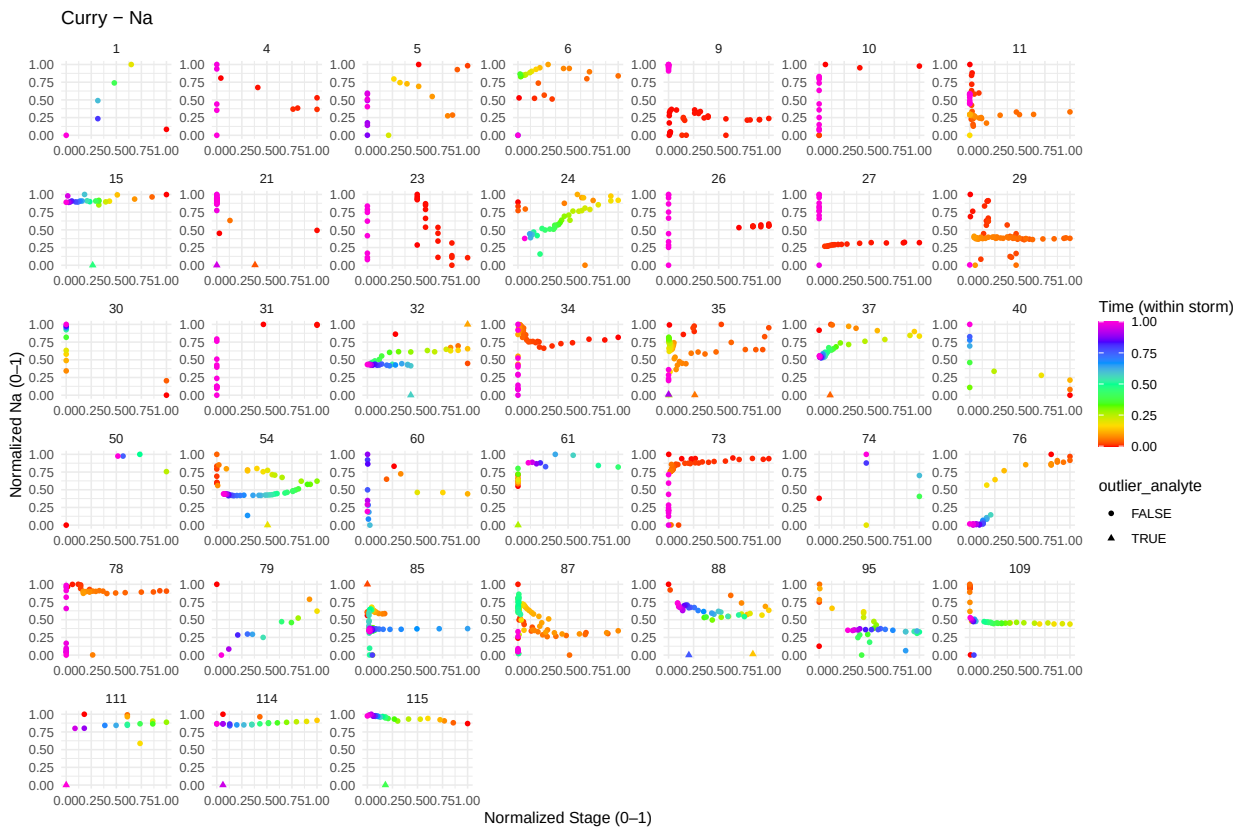
Curry − DOC



Curry − K




Curry − Mg
1
4
5
6
9
10
11
Normalized Mg (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE


Curry − Na
1
4
5
6
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10
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34
35
37
40
50
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60
61
73
74
76
78
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85
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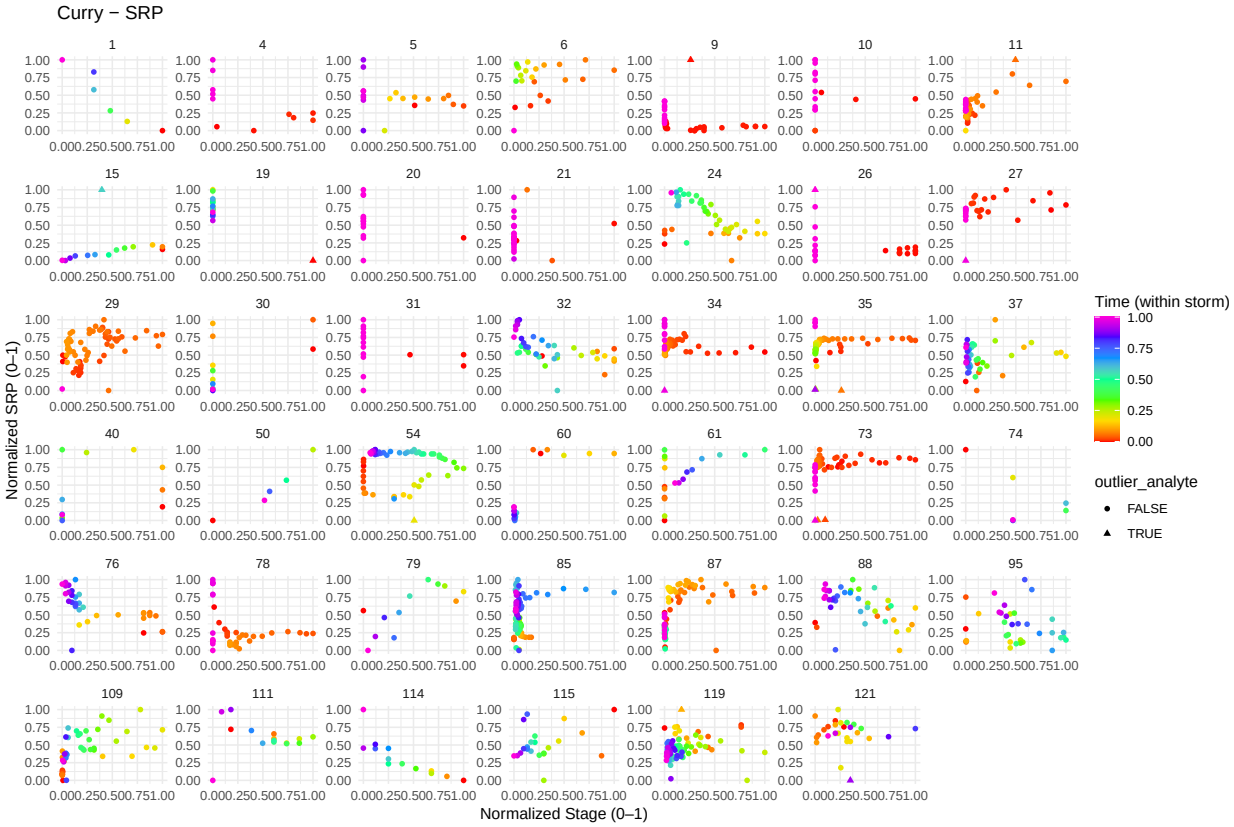
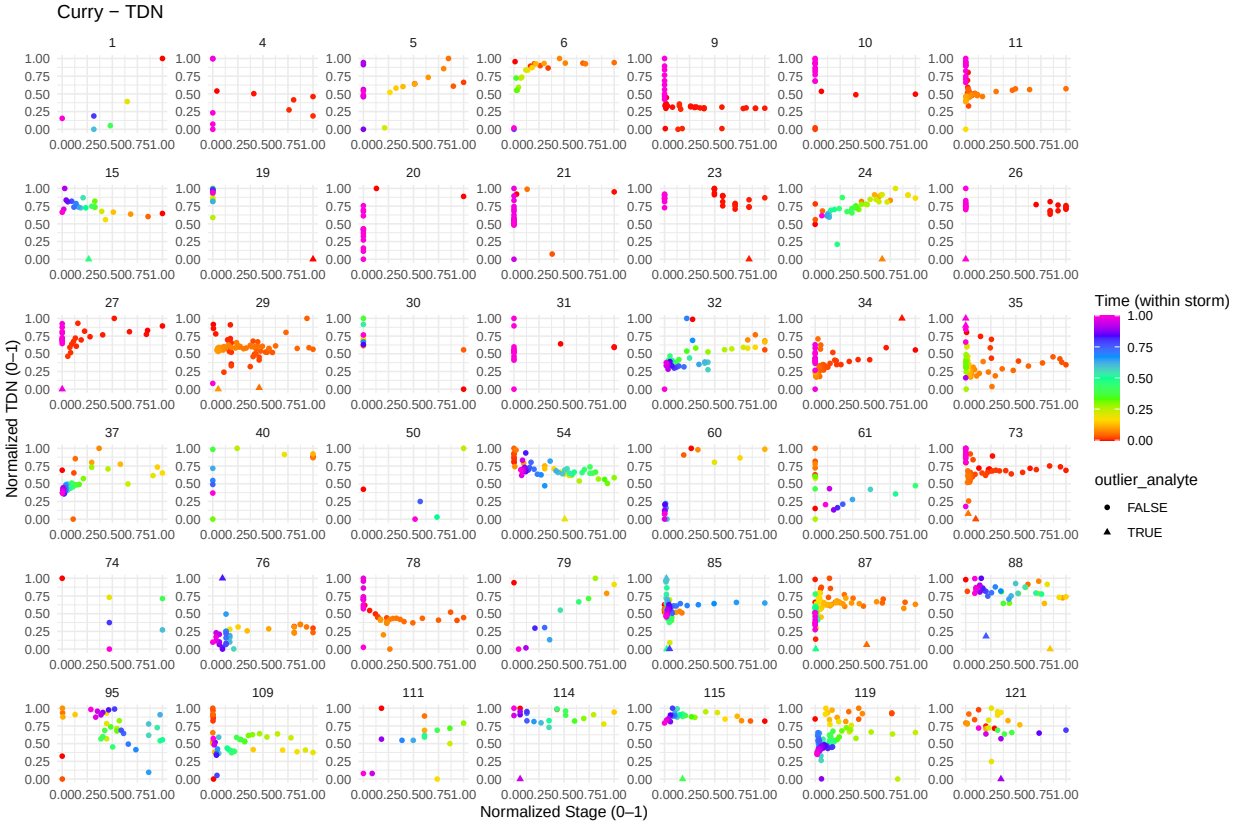
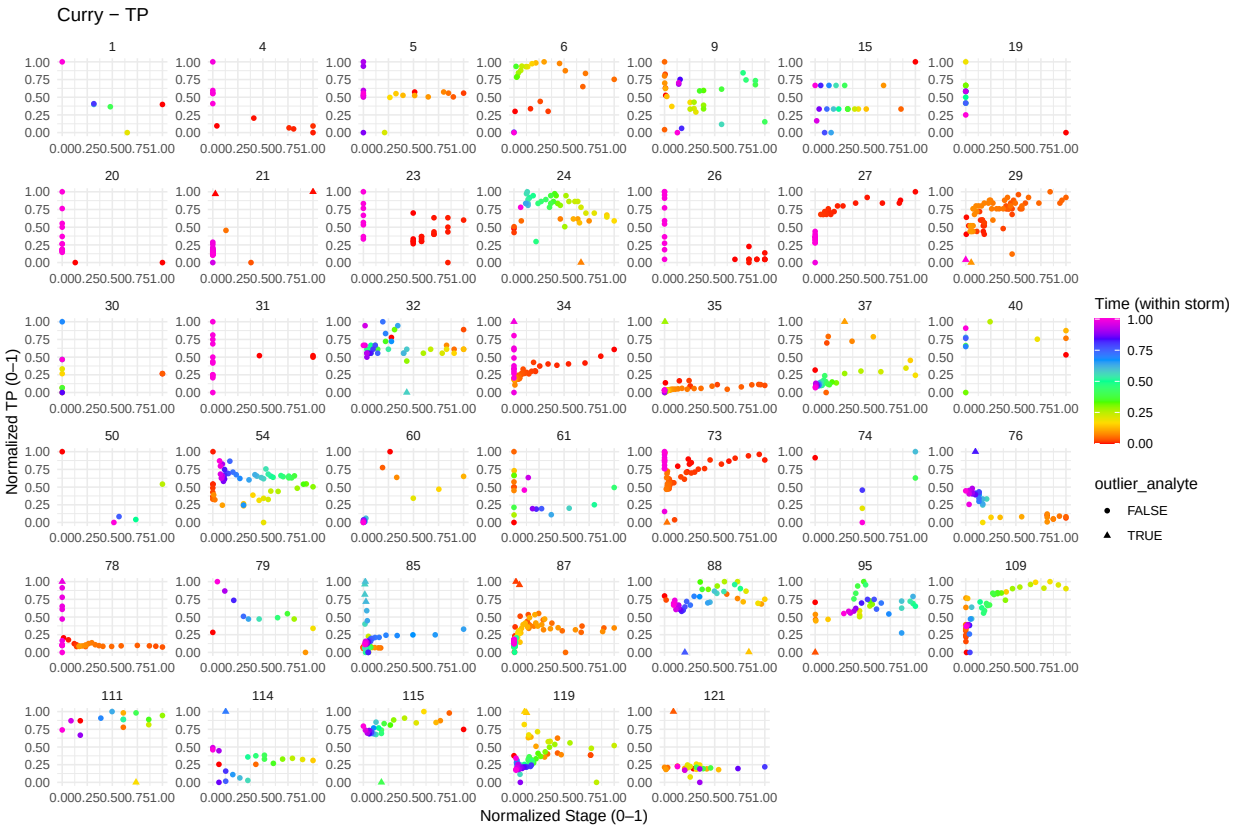
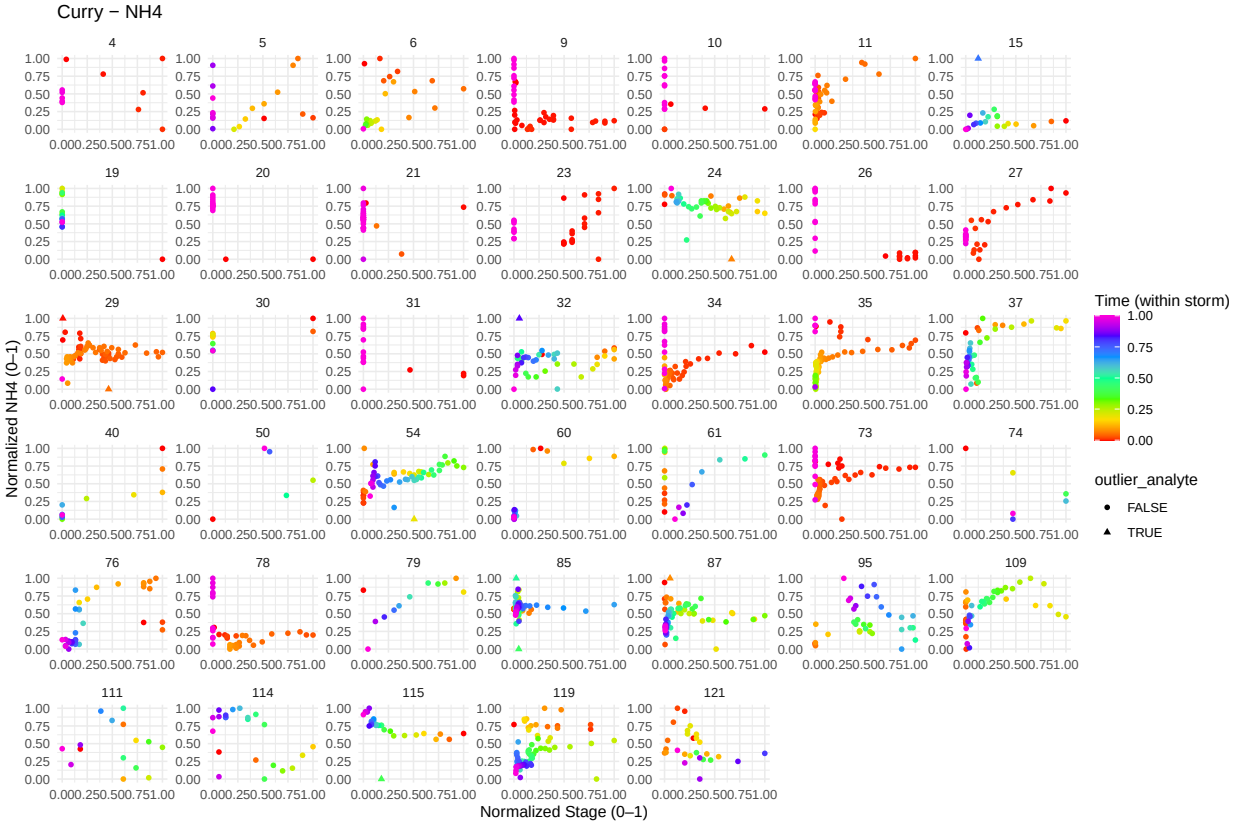
95
109
111
114
115
Normalized Na (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

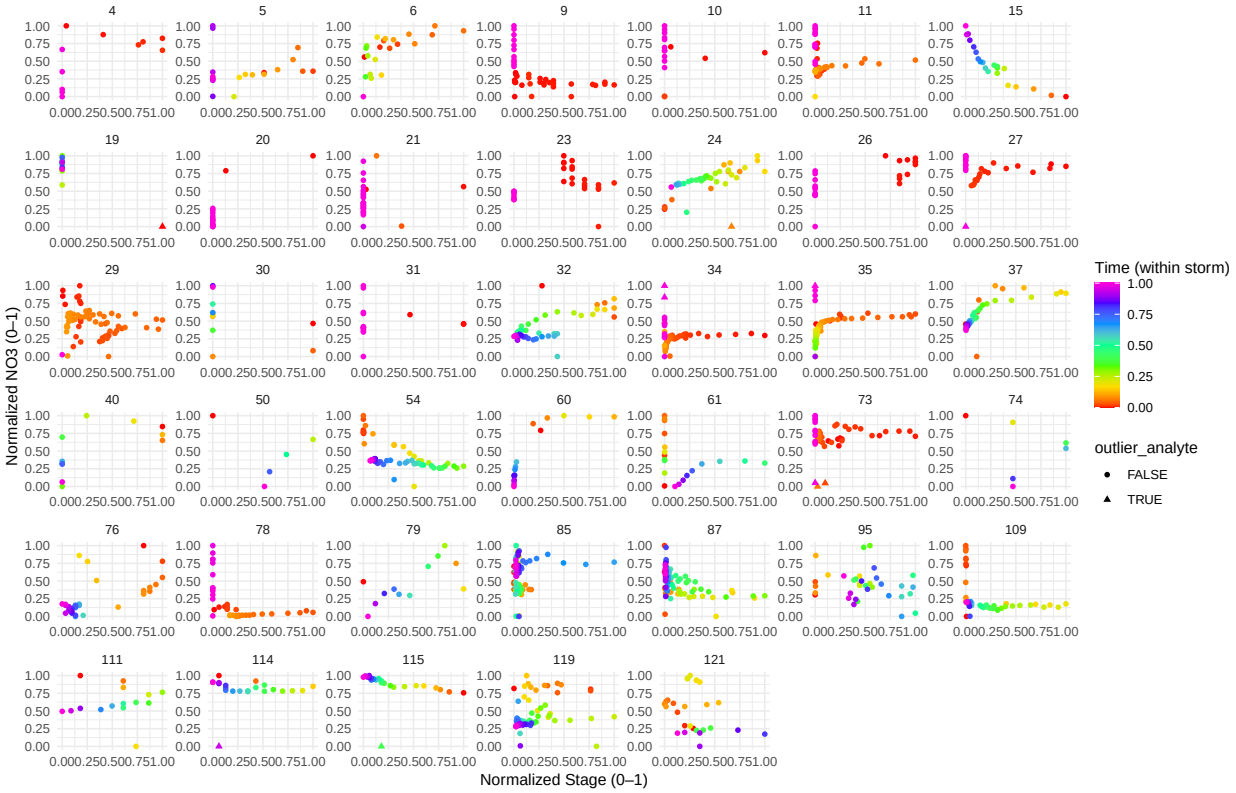

Curry − SRP
Normalized SRP (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Curry − TDN
Normalized TDN (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

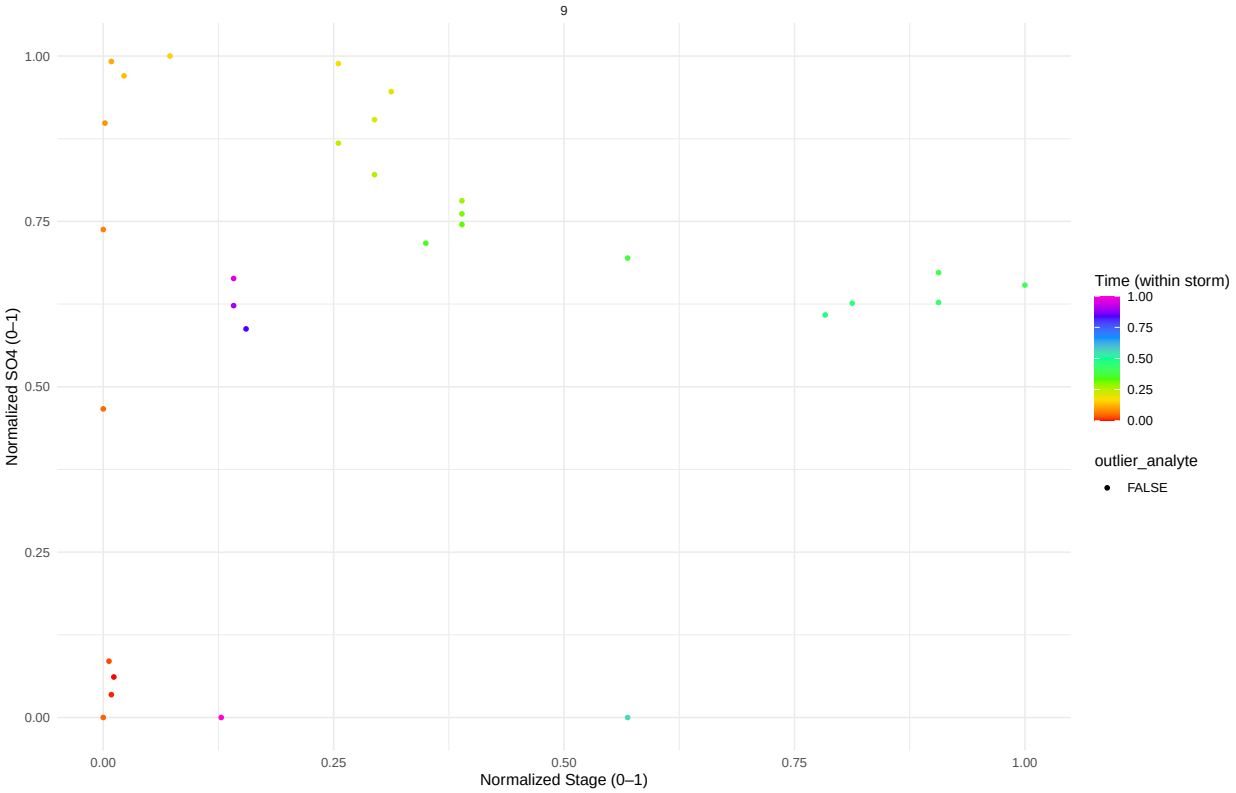

Curry − TP
Normalized TP (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

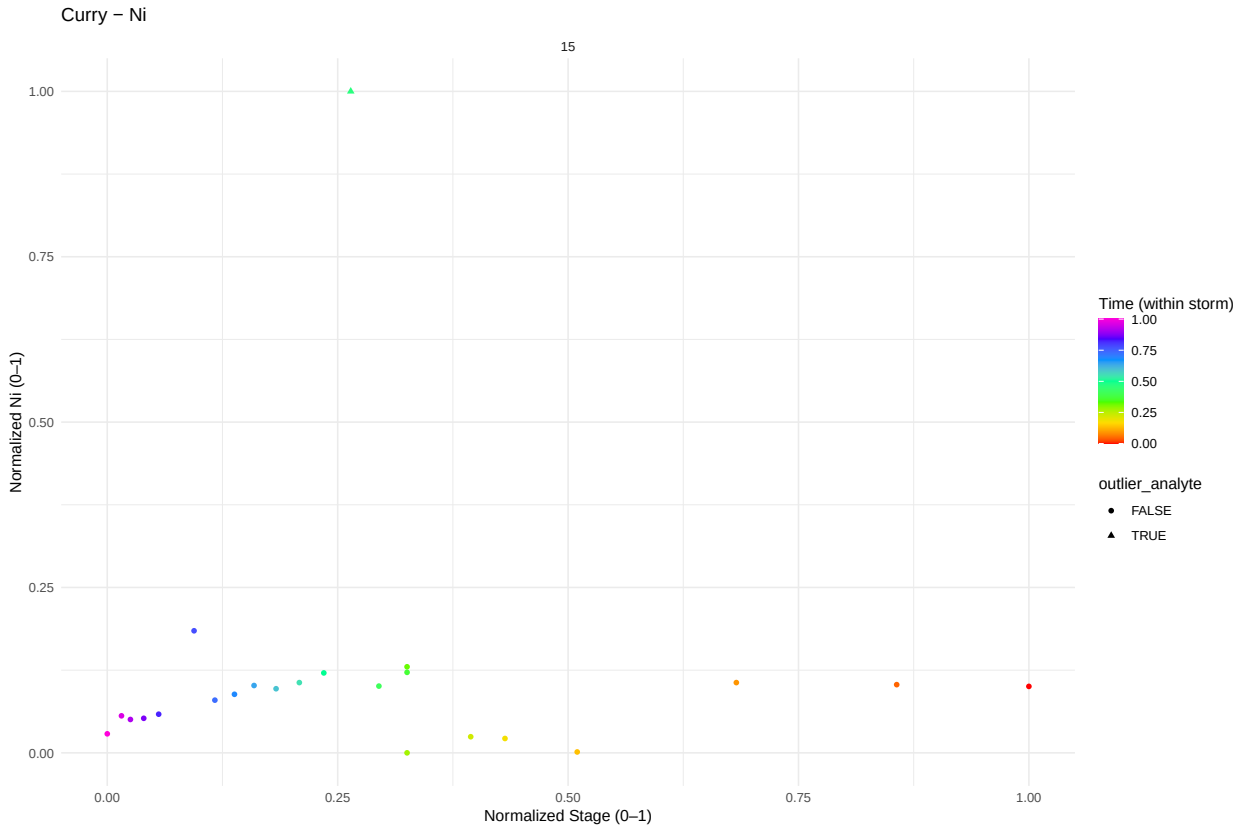

Curry − NH4
Normalized NH4 (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

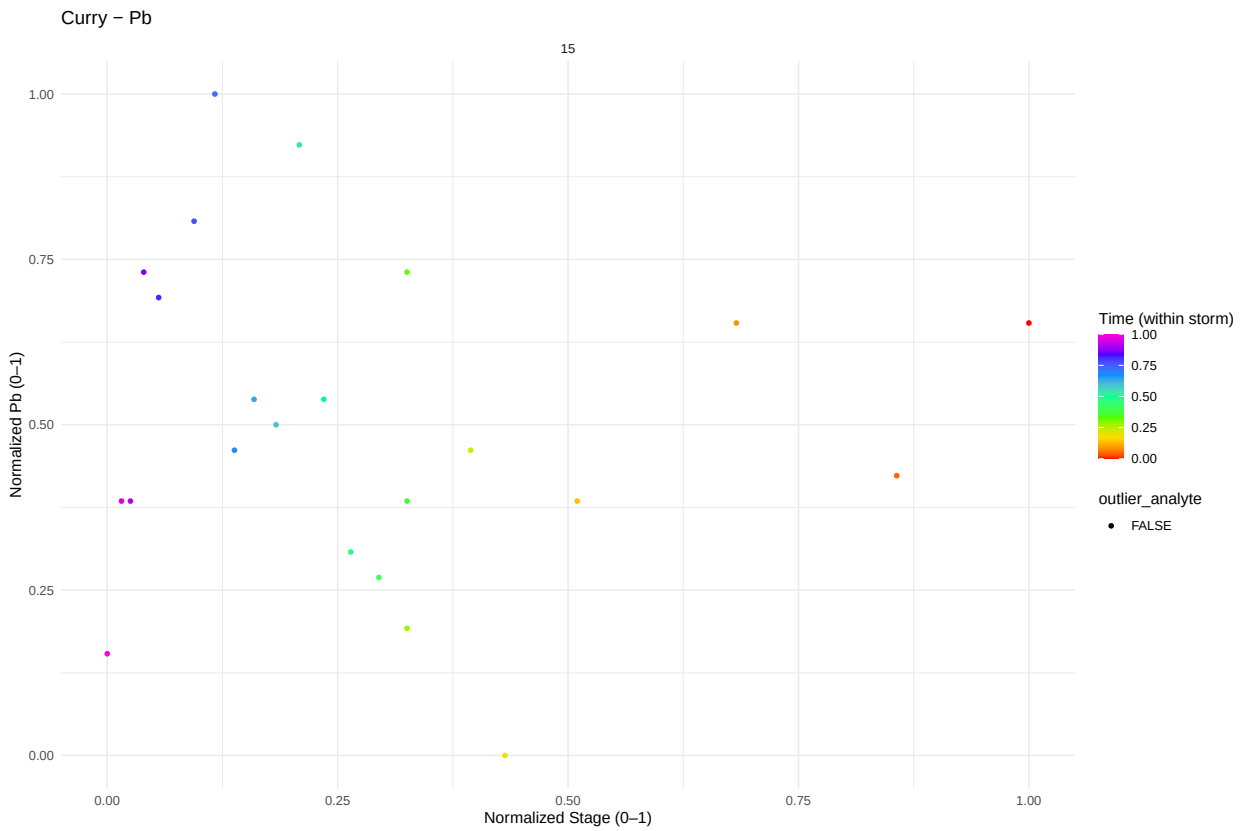
Curry − NO3

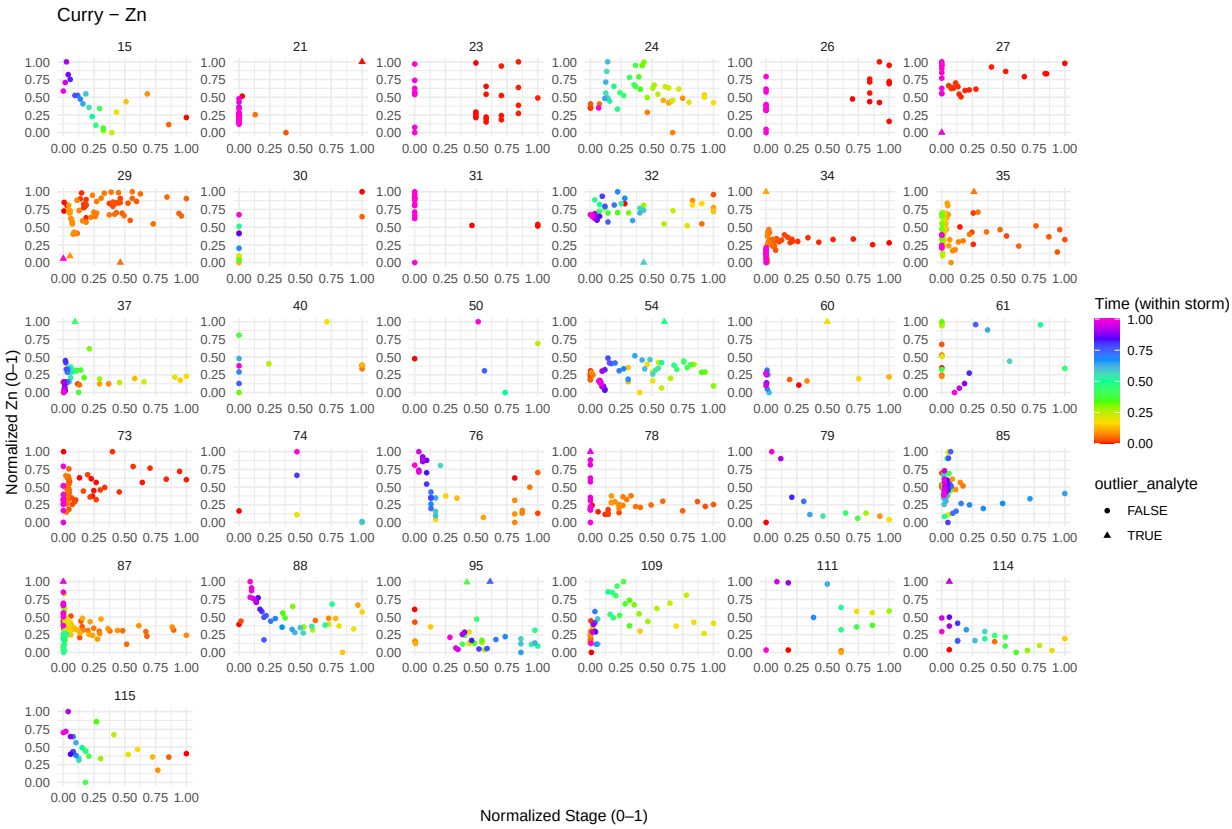


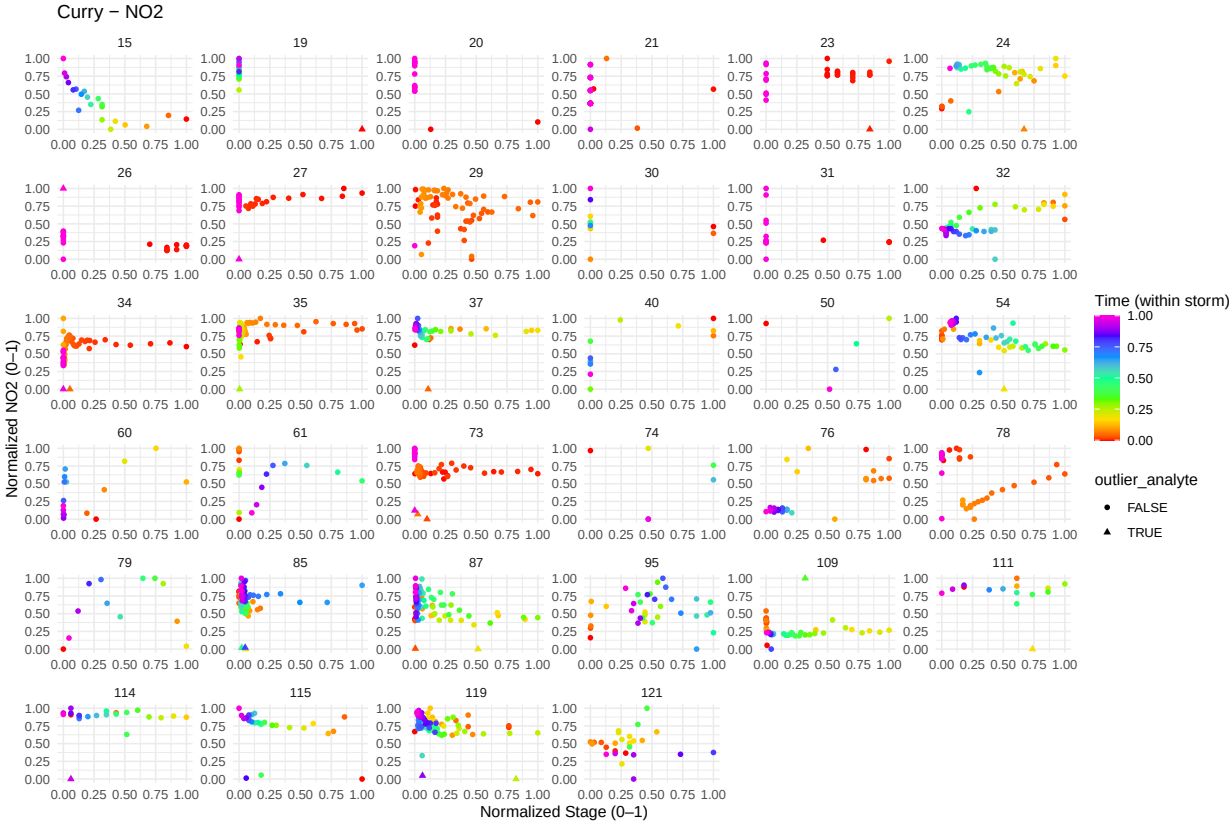
Curry − SO4

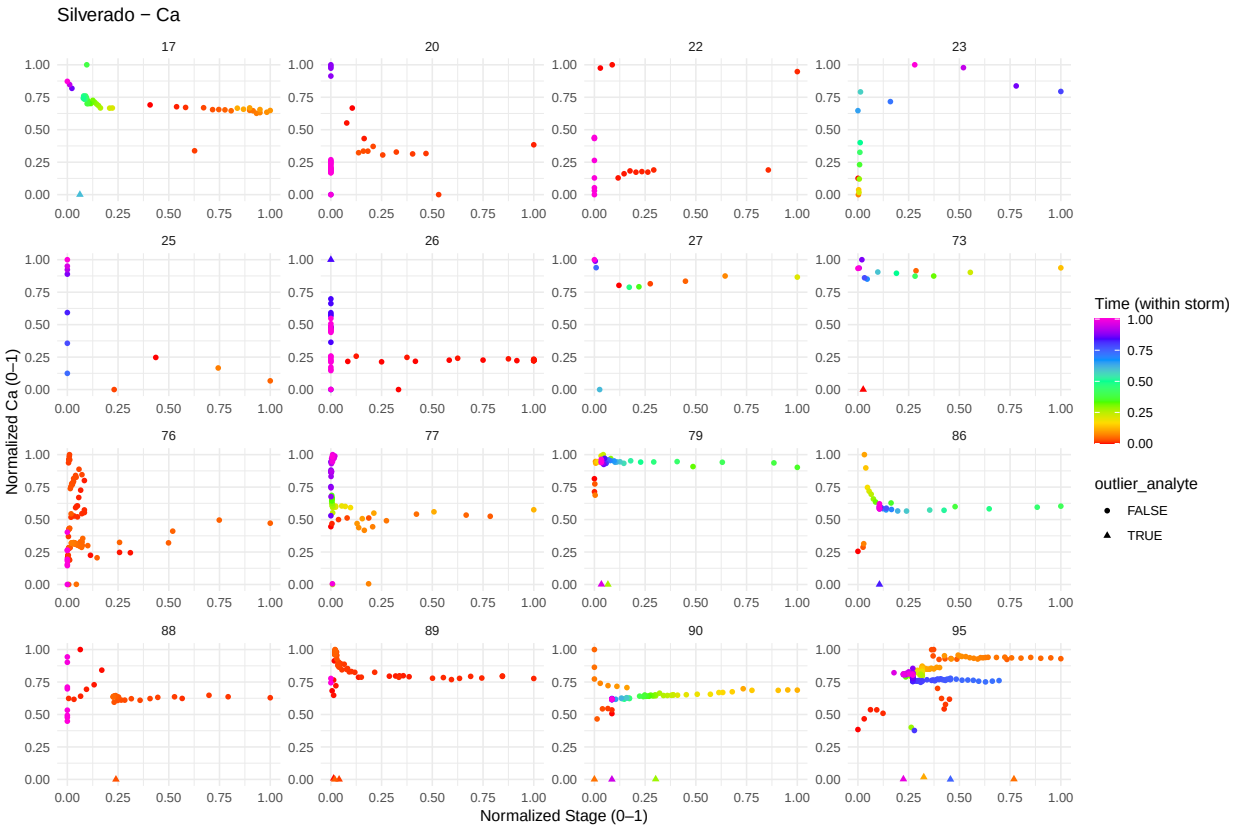



Curry − Ni
15
Normalized Ni (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Curry − Pb
15
Normalized Pb (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE


Curry − Zn
Normalized Zn (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Curry − NO2
Normalized NO2 (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − Ca
17
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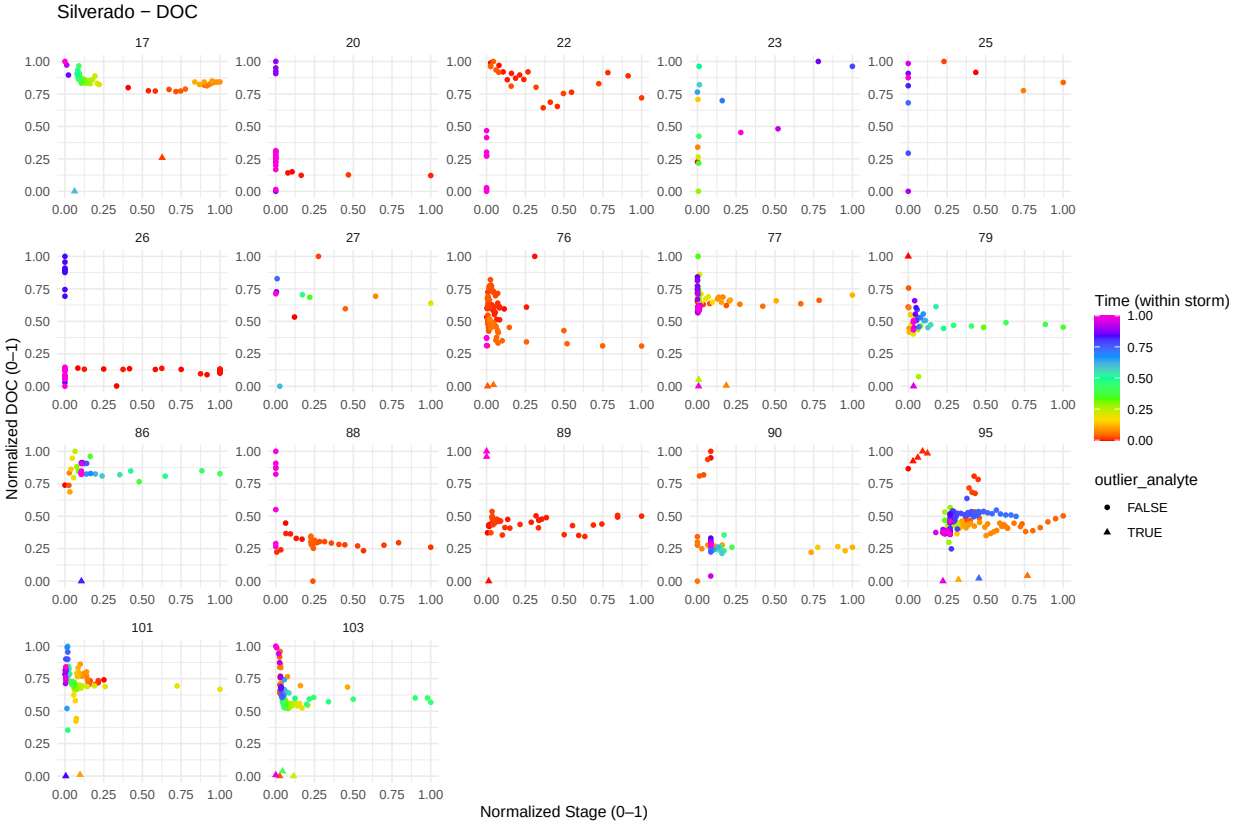
89
90
95
Normalized Ca (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − DOC
17
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23
25
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76
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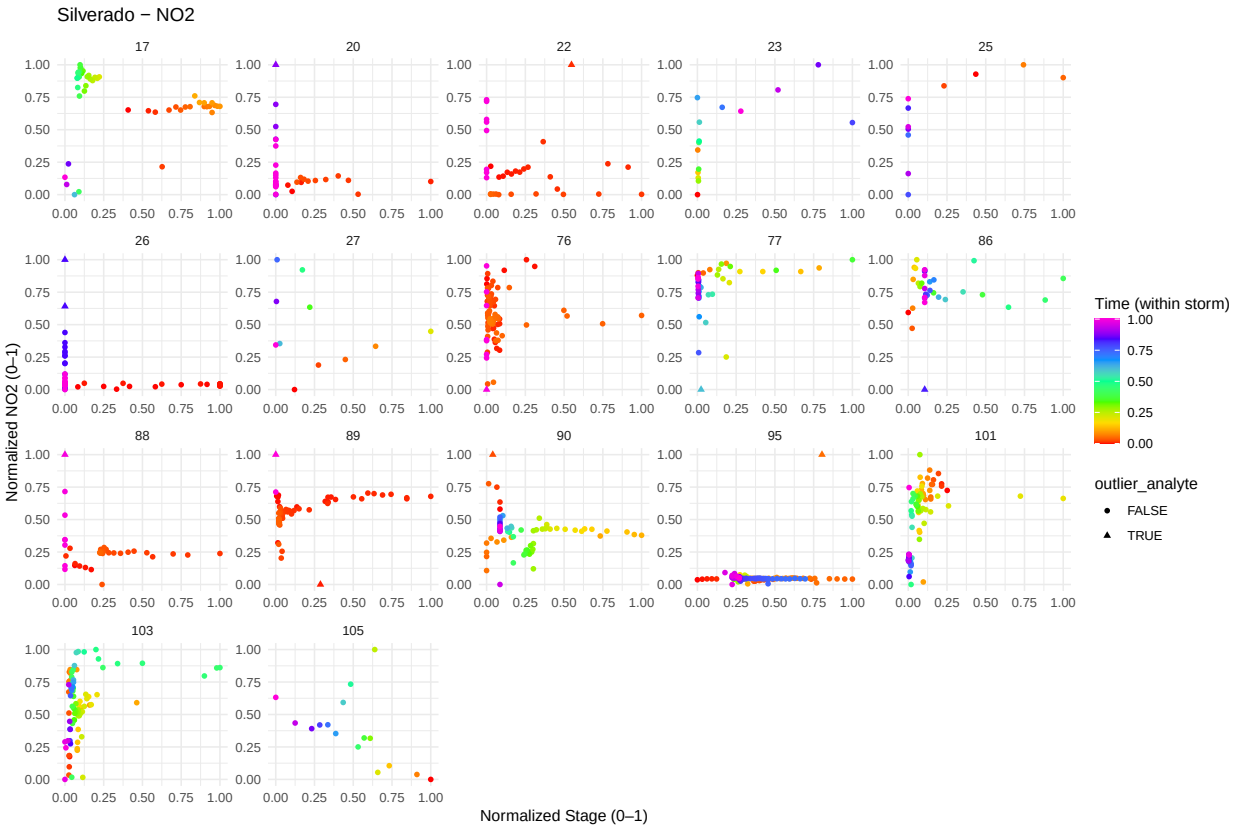
88
89
90
95
101
103
Normalized DOC (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − NO2
17
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23
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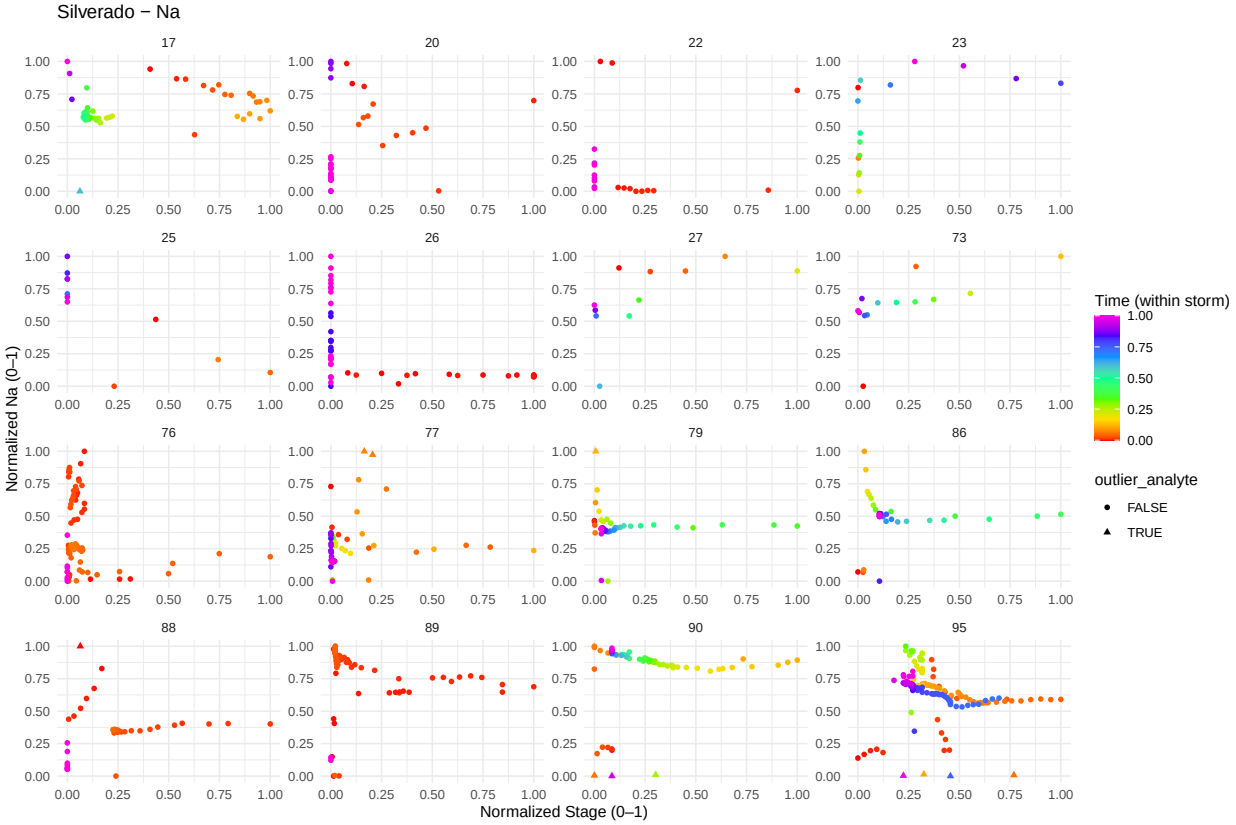
26
27
76
77
86
88
89
90
95
101
103
105
Normalized NO2 (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − Na
17
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86
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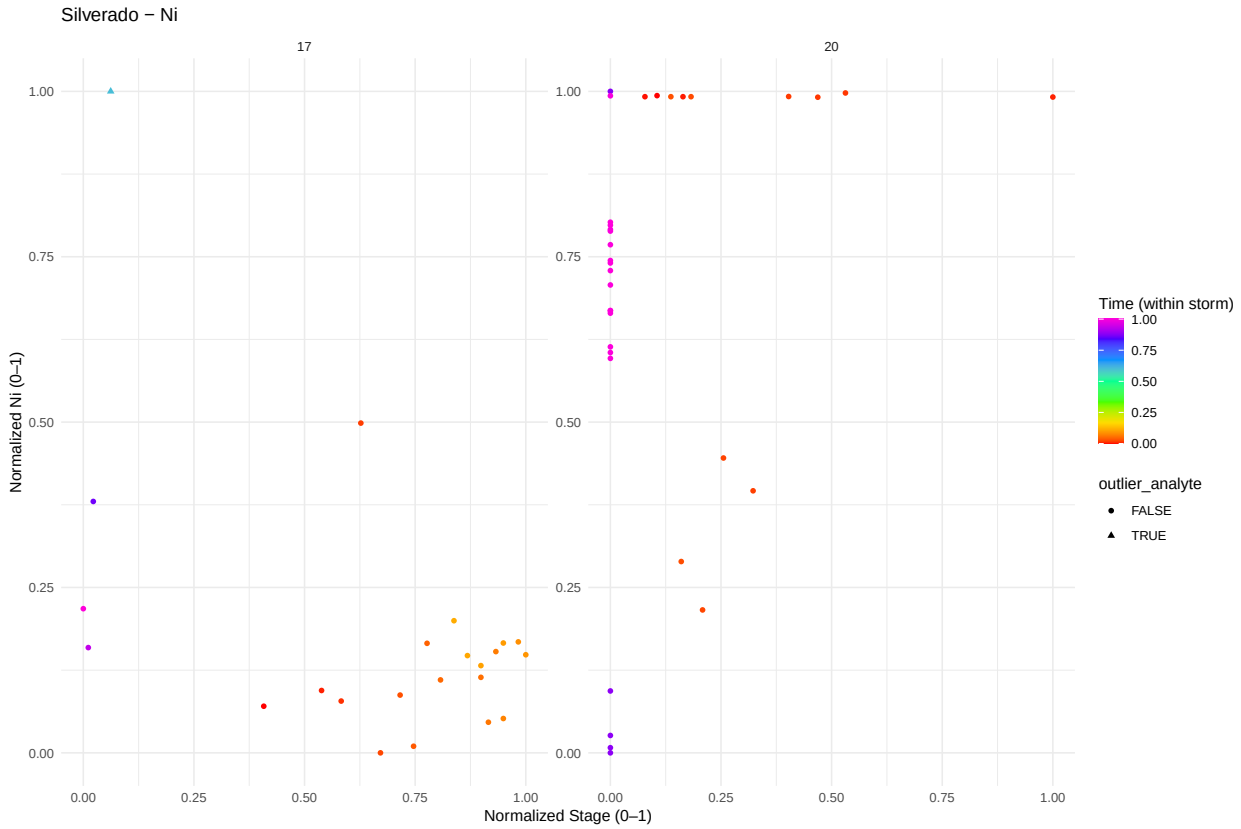
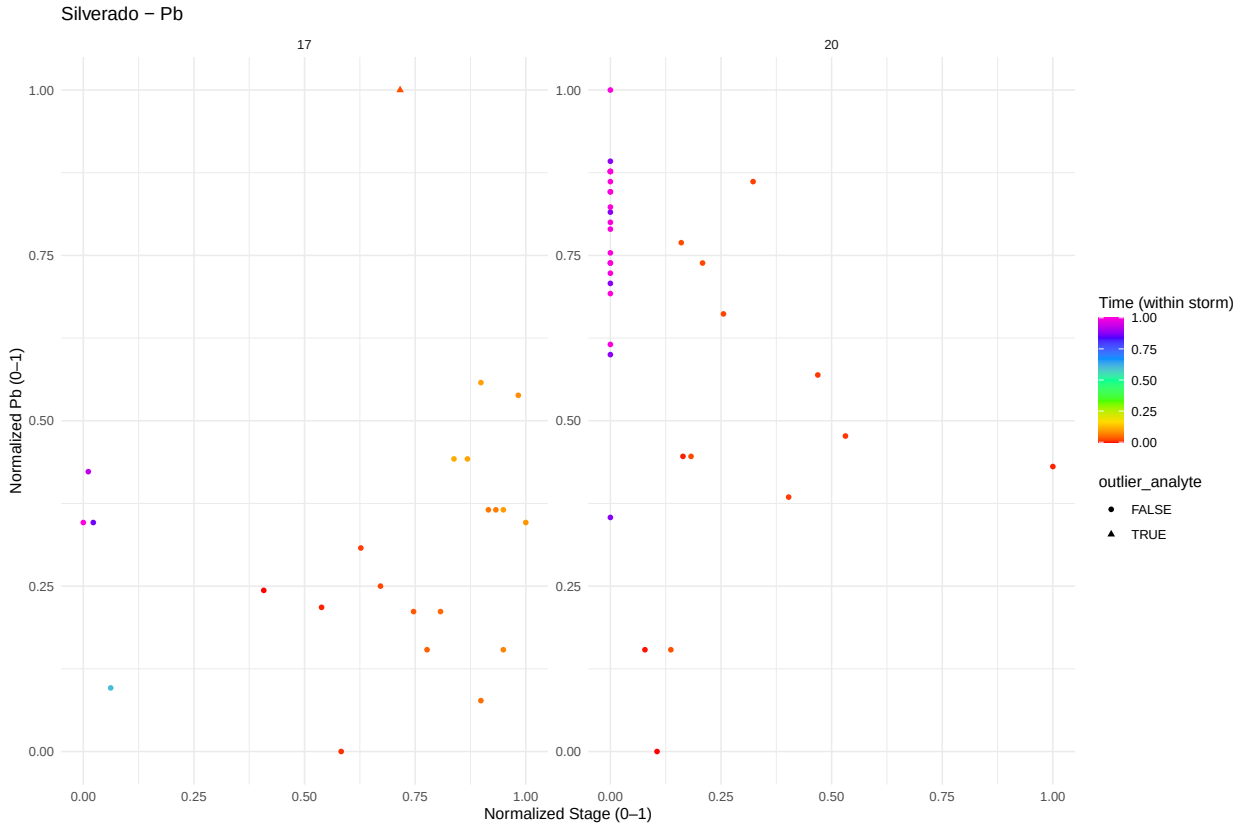
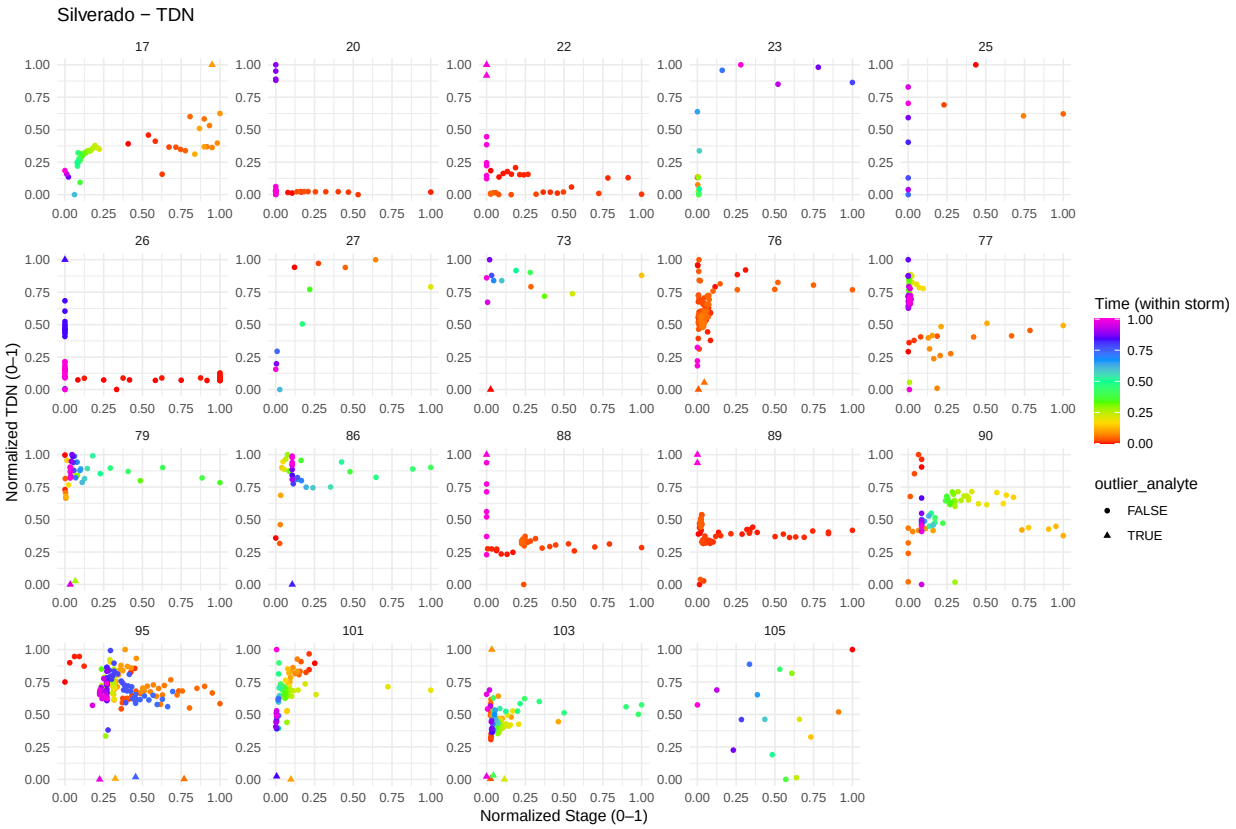
89
90
95
Normalized Na (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − Ni
17
20
Normalized Ni (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − Pb
17
20
Normalized Pb (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − TDN
17
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27
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86
88
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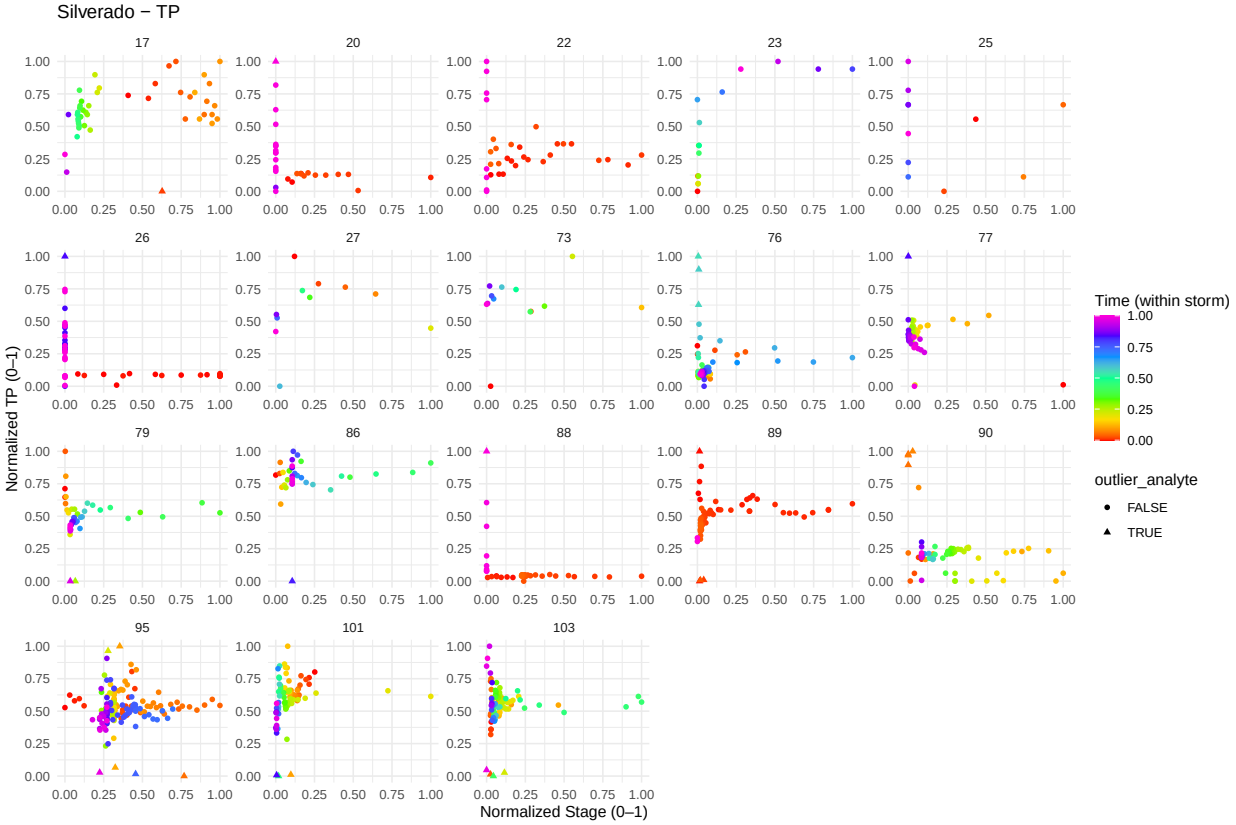
90
95
101
103
105
Normalized TDN (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − TP
17
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89
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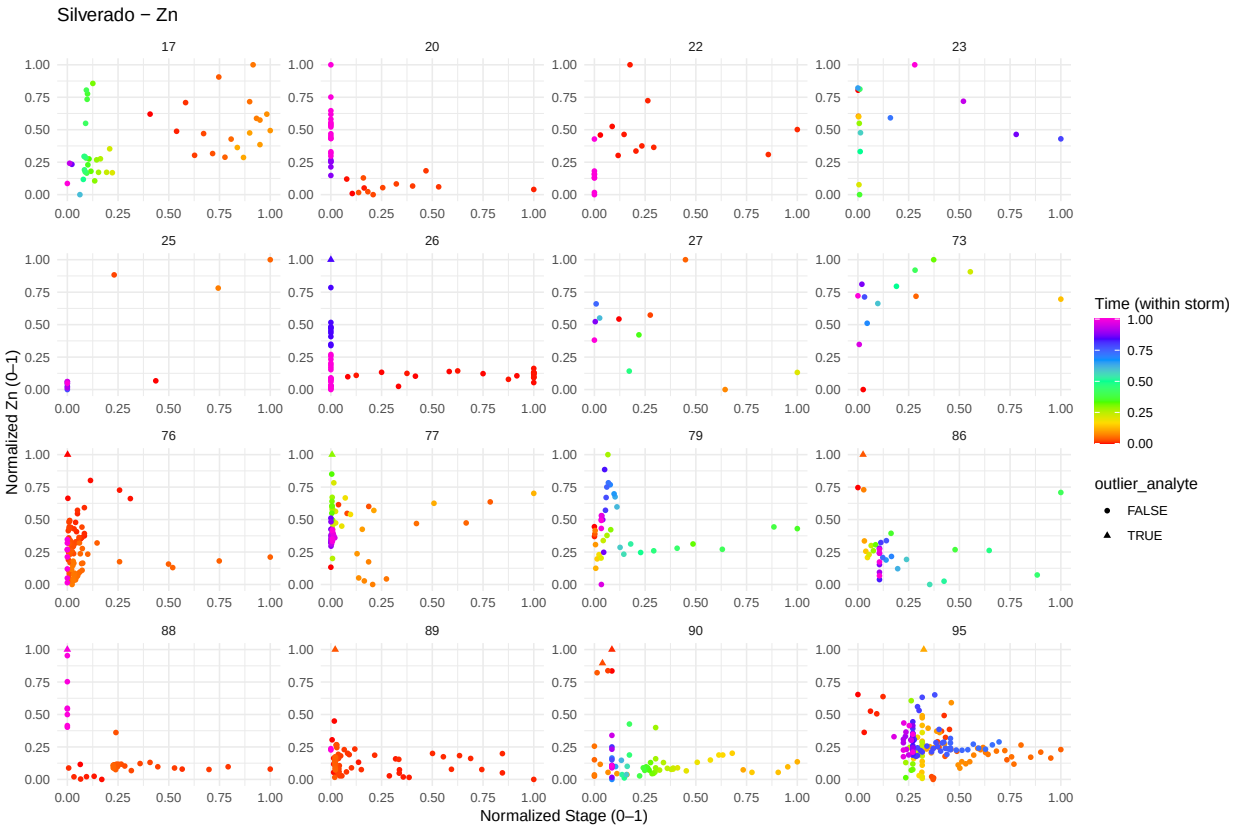
95
101
103
Normalized TP (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − Zn
17
20
22
23
25
26
27
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76
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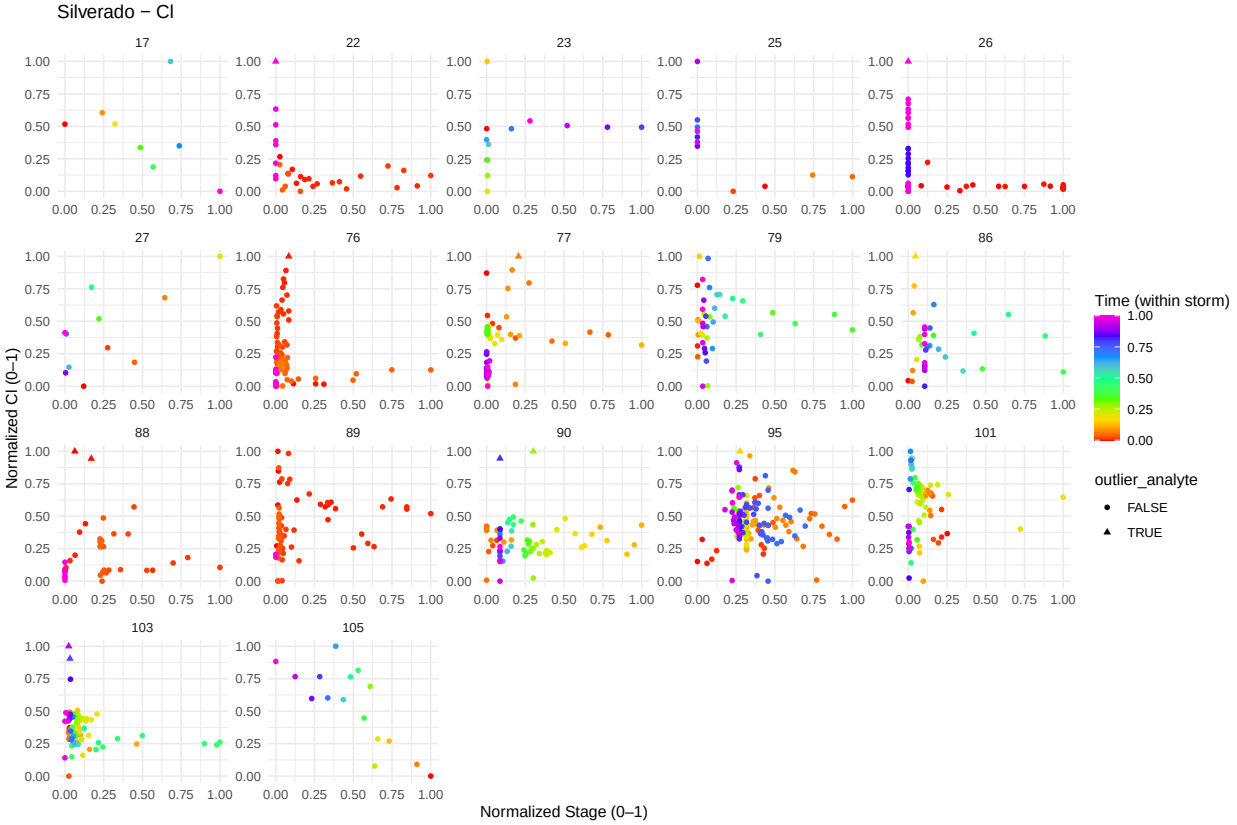
79
86
88
89
90
95
Normalized Zn (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − Cl
17
22
23
25
26
27
76
77
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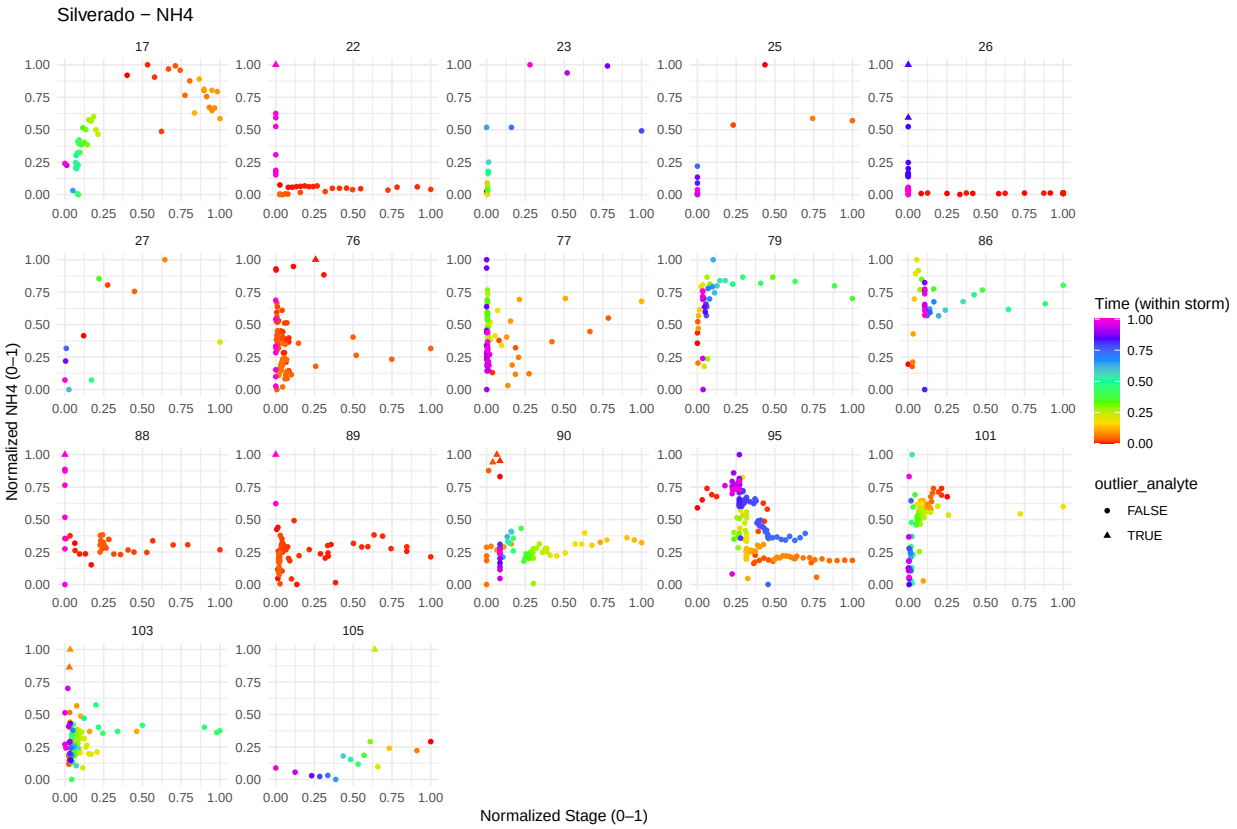
86
88
89
90
95
101
103
105
Normalized Cl (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − NH4
17
22
23
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26
27
76
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86
88
89
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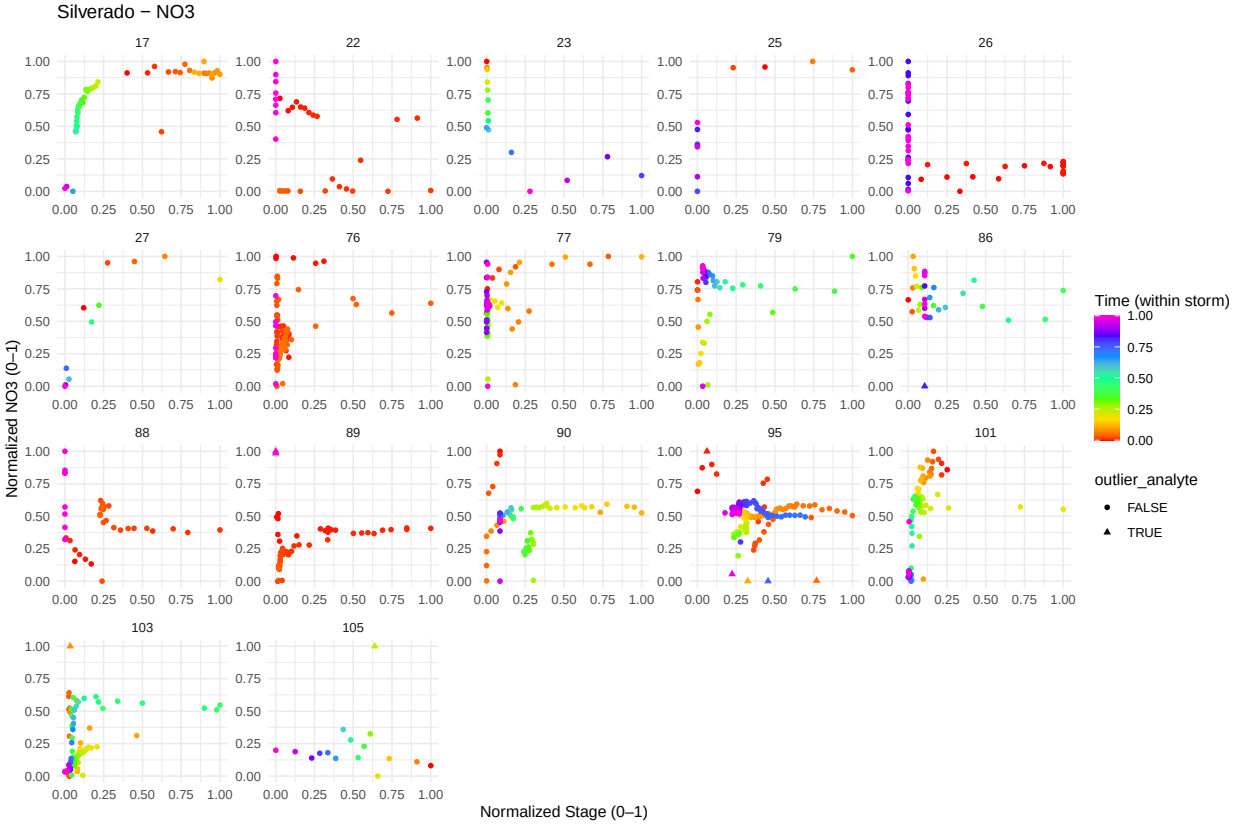
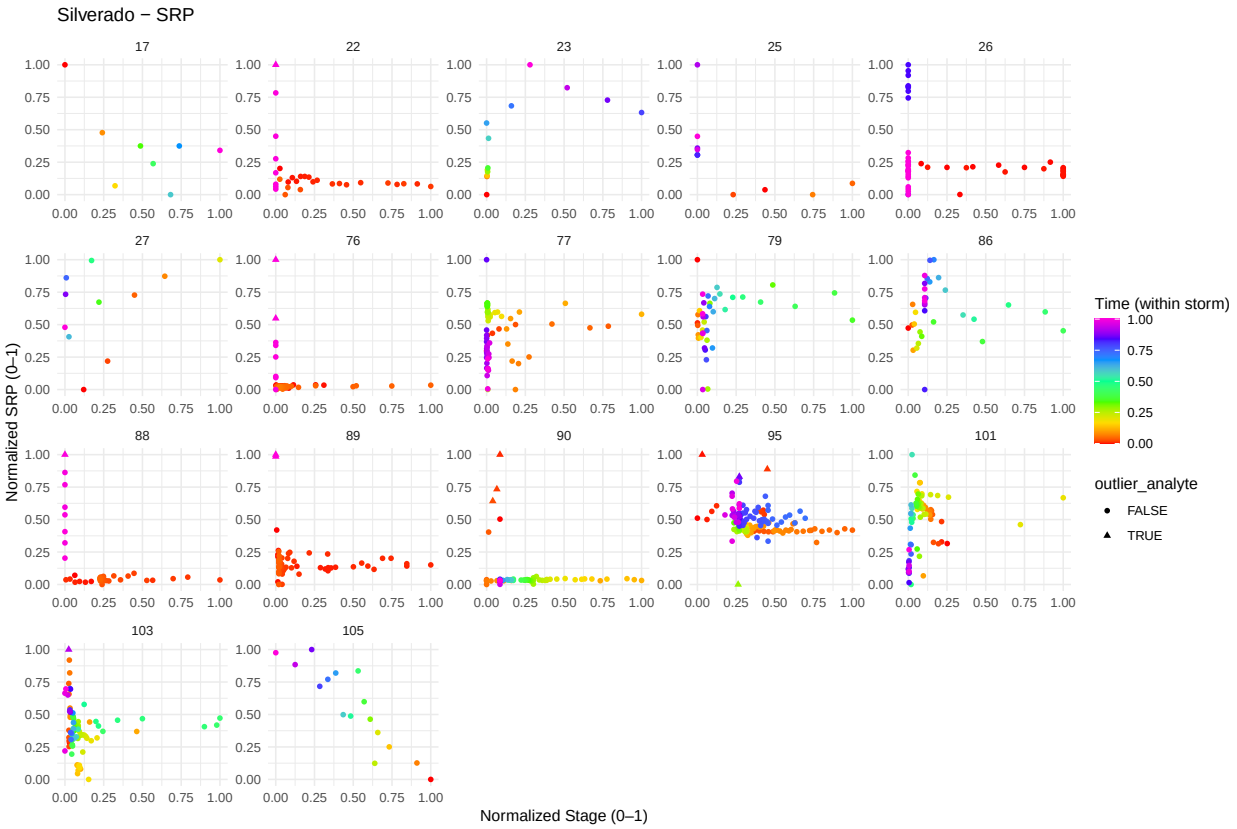
101
103
105
Normalized NH4 (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − NO3
17
22
23
25
26
27
76
77
79
86
88
89
90
95
101
103
105
Normalized NO3 (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE


Silverado − SRP
17
22
23
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26
27
76
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86
88
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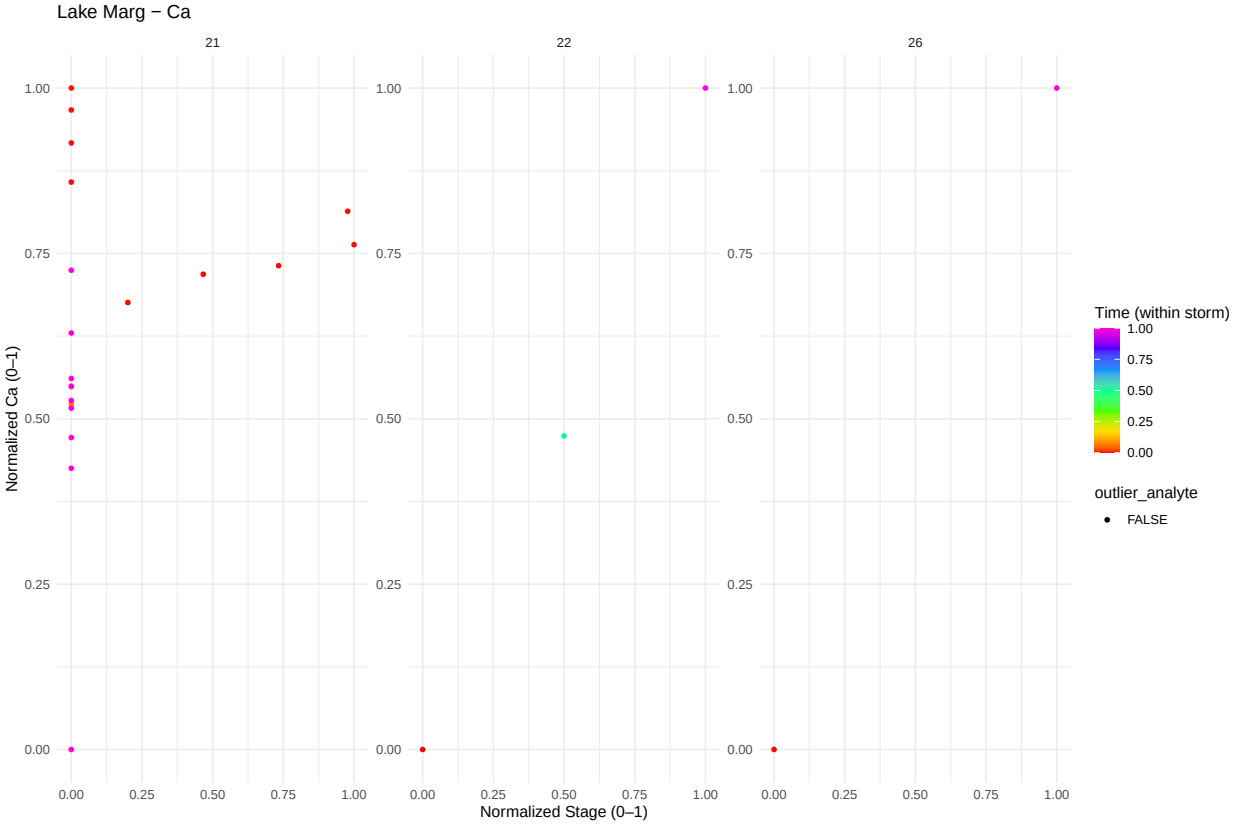
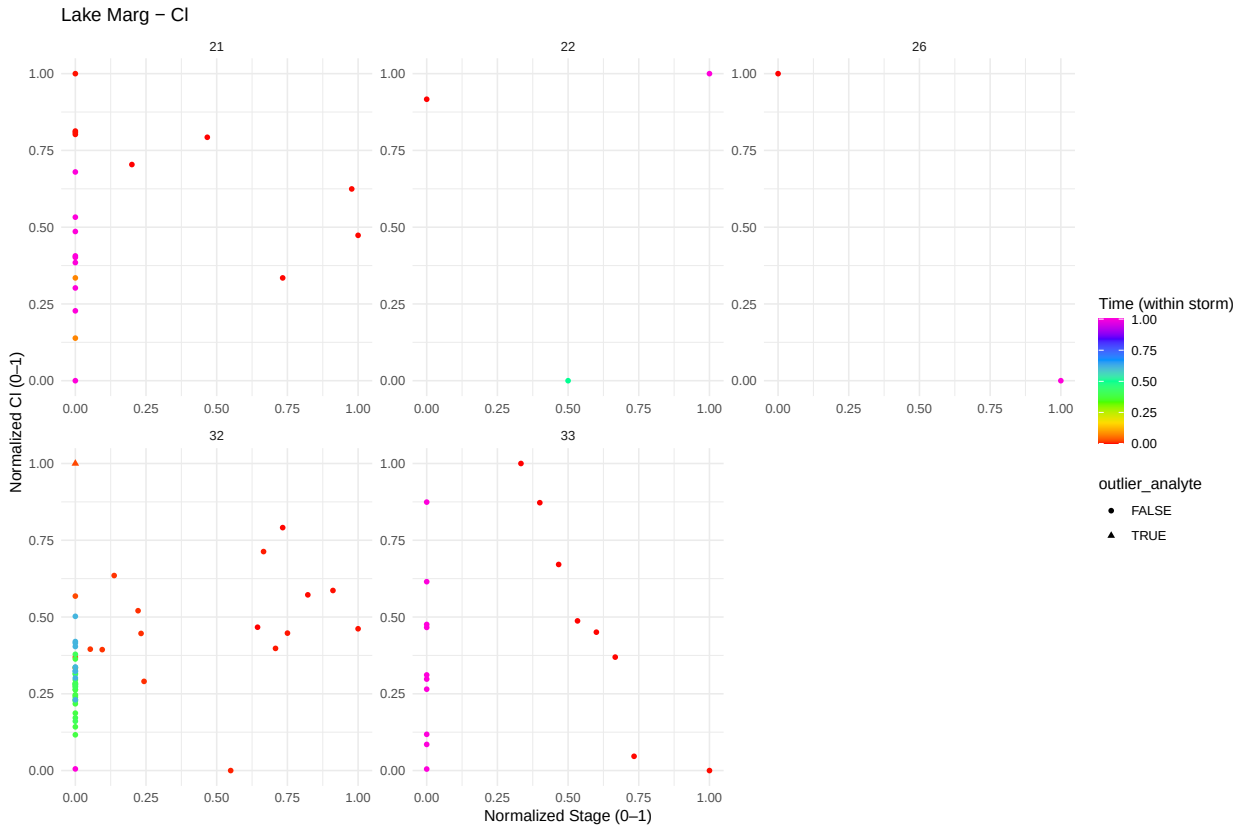
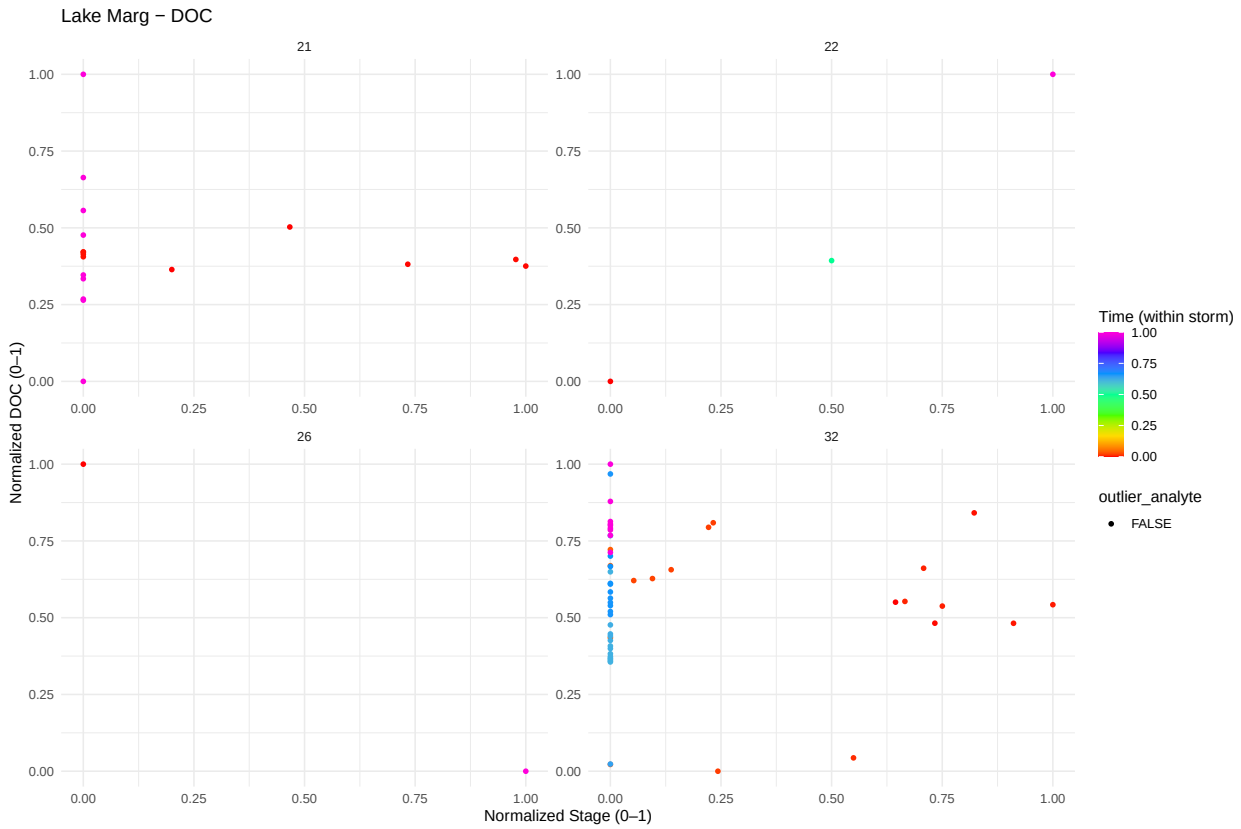
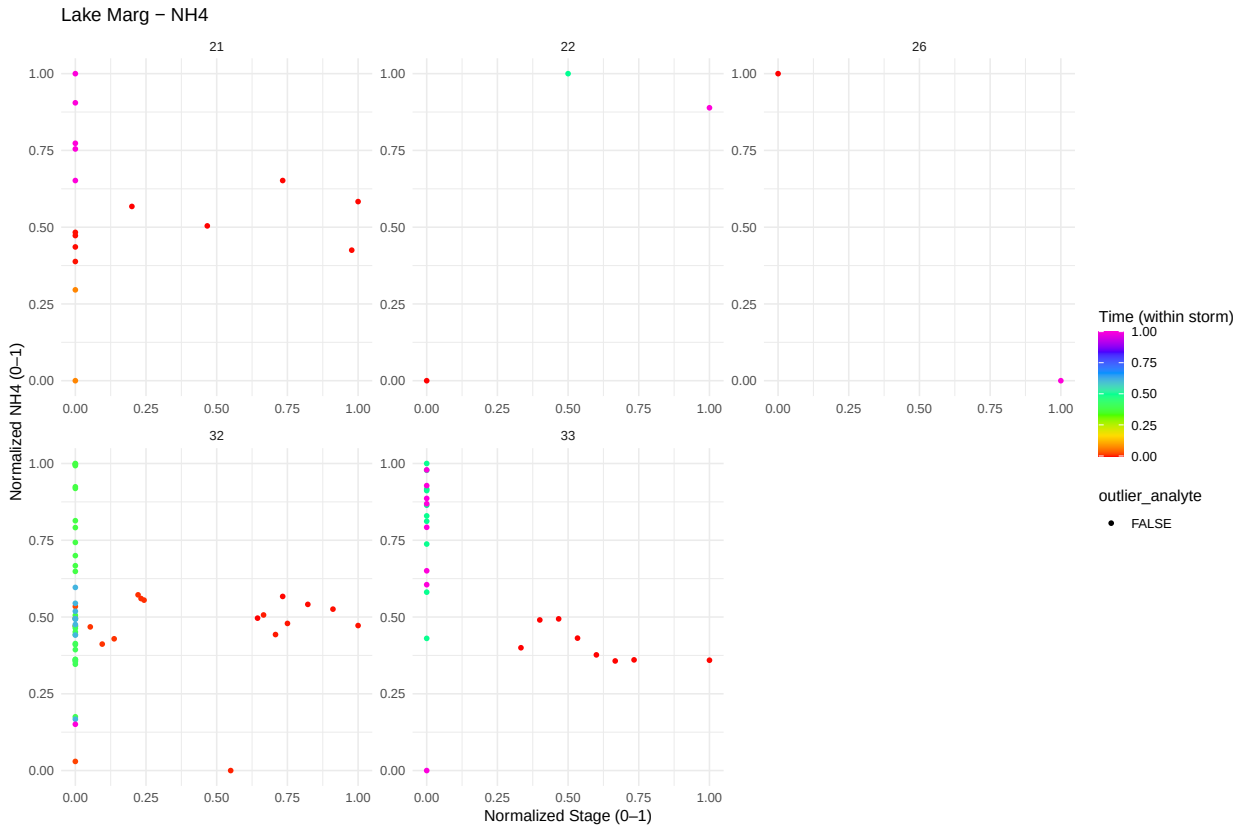
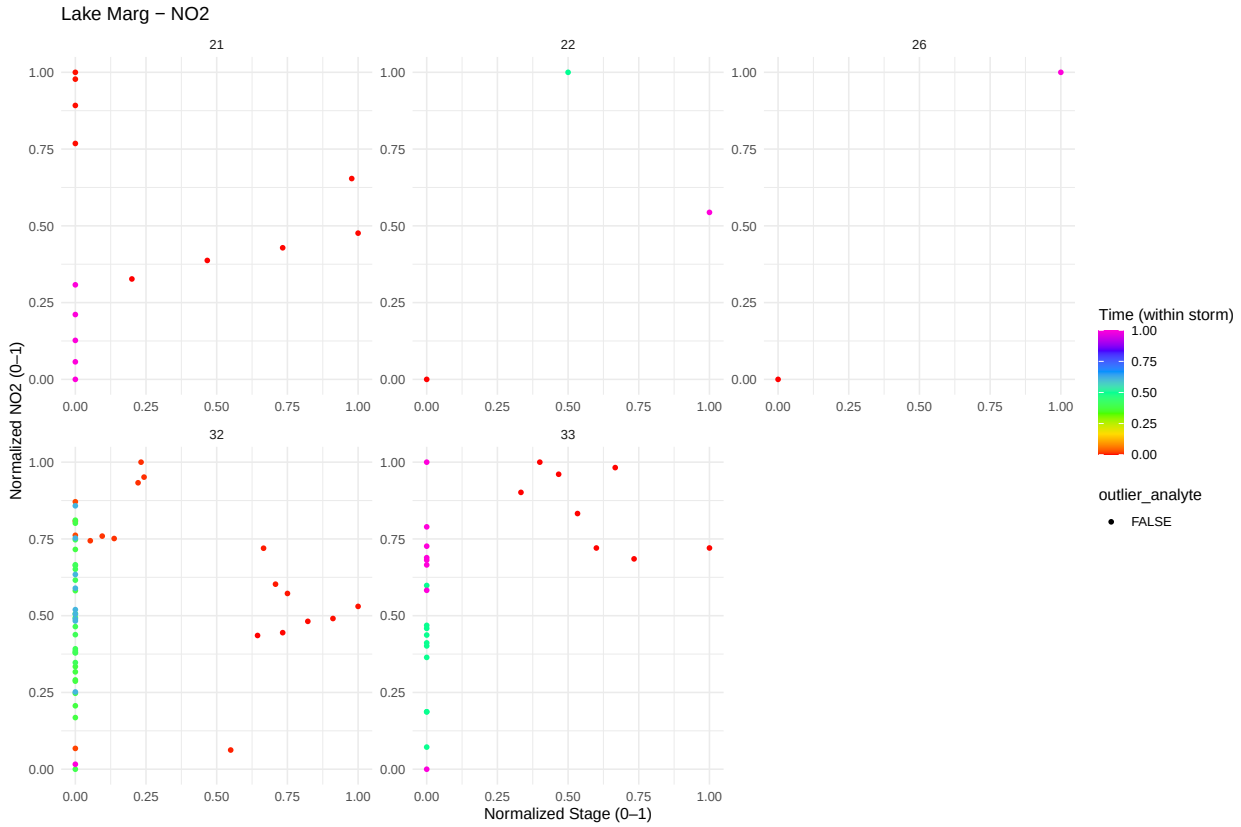
90
95
101
103
105
Normalized SRP (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

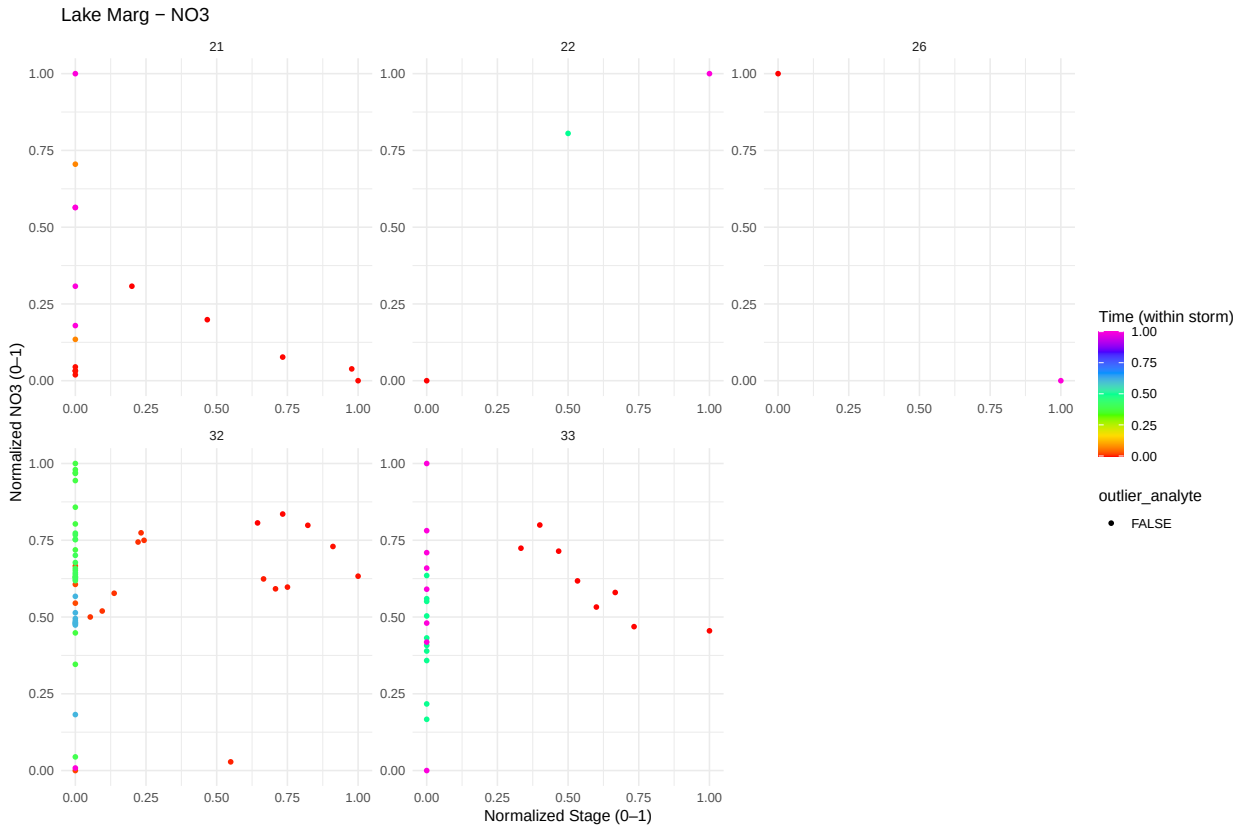

Lake Marg − Ca
21
22
26
Normalized Ca (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE

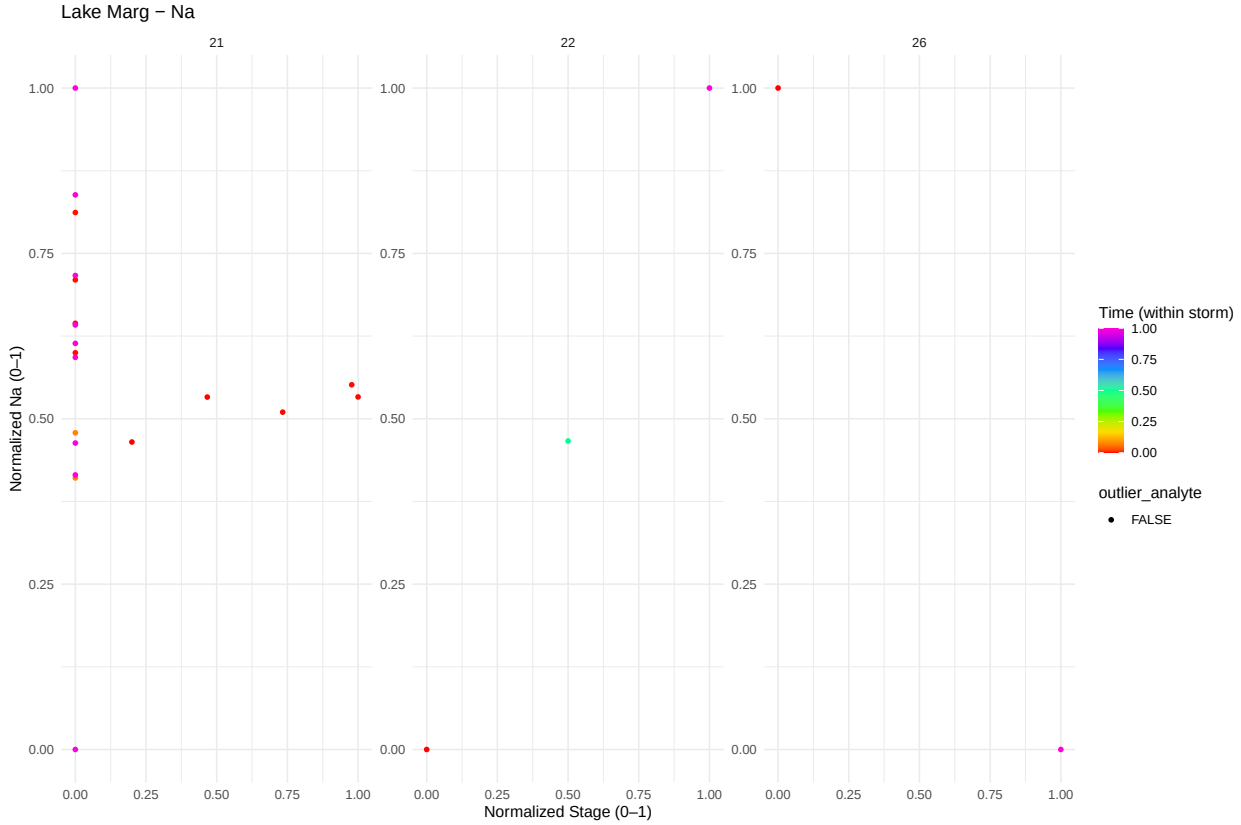

Lake Marg − Cl
21
22
26
32
33
Normalized Cl (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

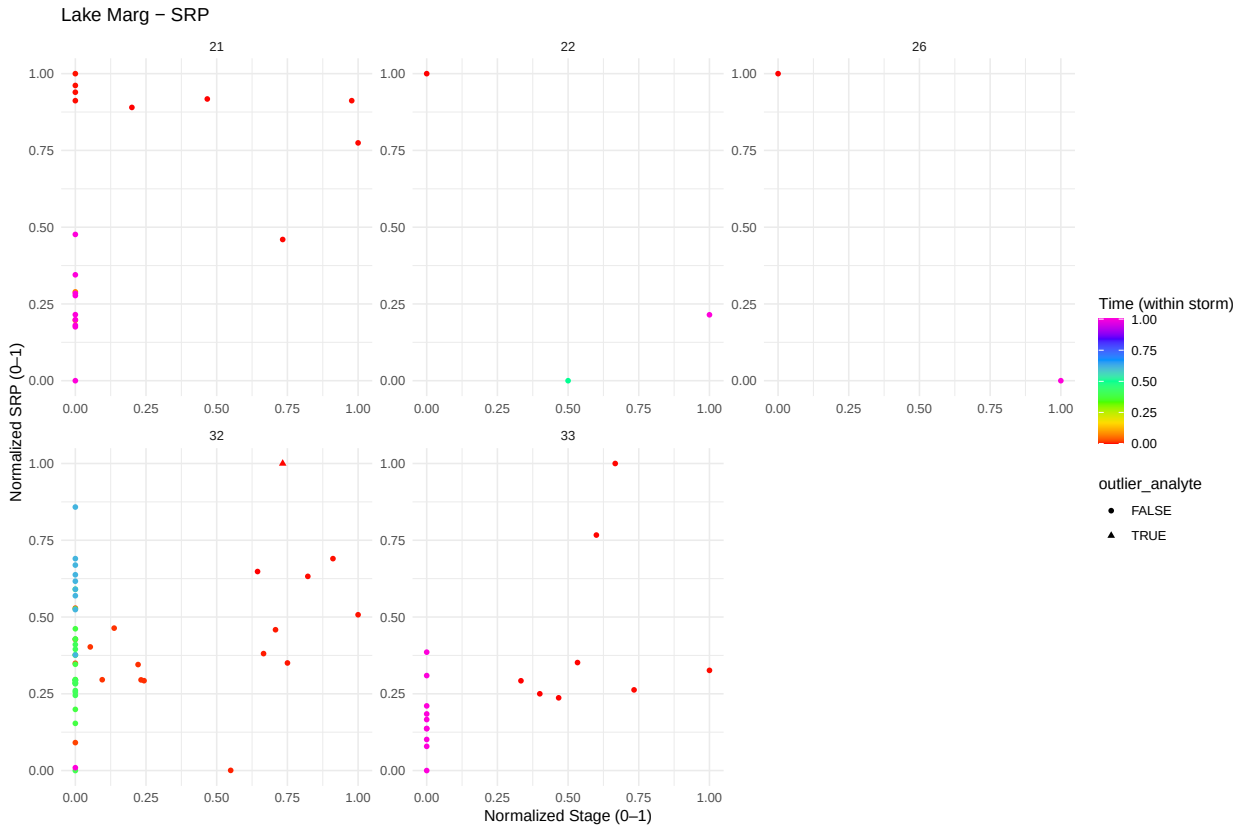

Lake Marg − DOC
21
22
26
32
Normalized DOC (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE

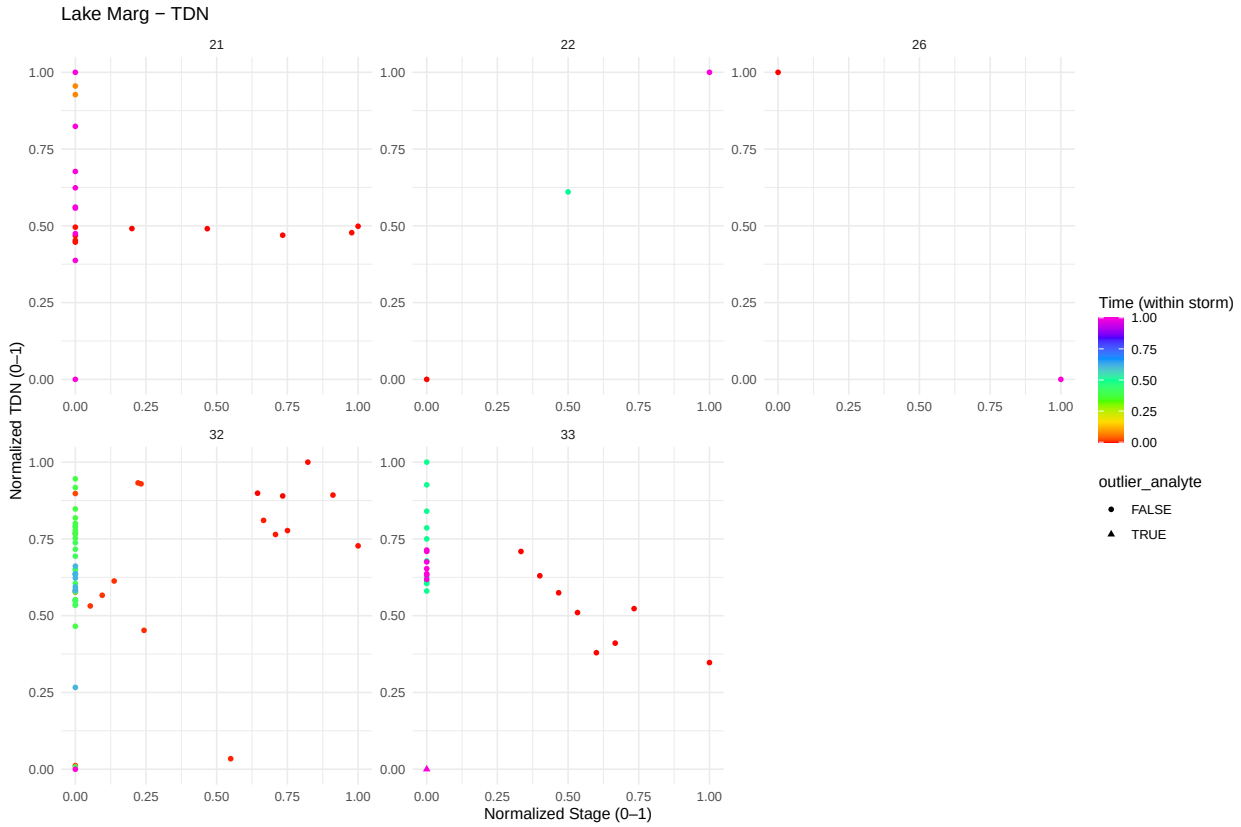

Lake Marg − NH4
21
22
26
32
33
Normalized NH4 (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE

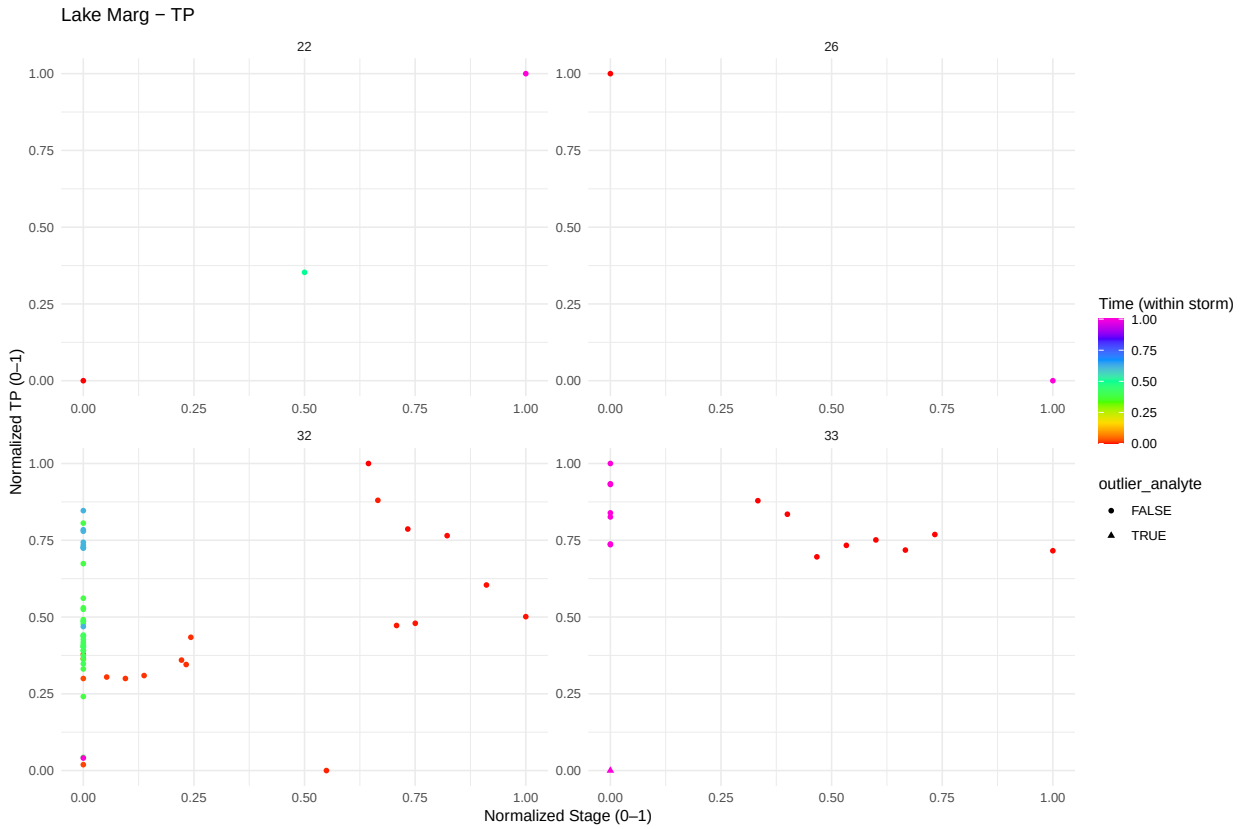

Lake Marg − NO2
21
22
26
32
33
Normalized NO2 (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE

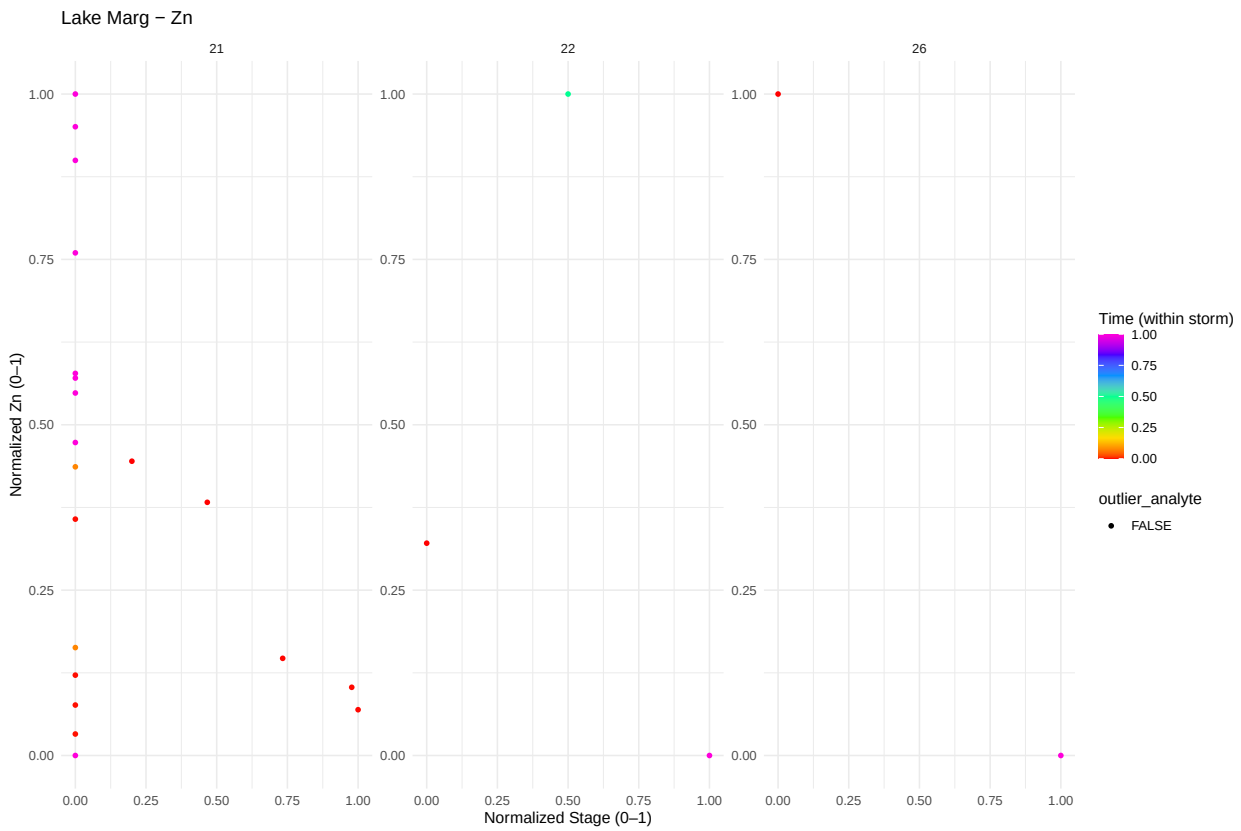

Lake Marg − NO3
21
22
26
32
33
Normalized NO3 (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE


Lake Marg − Na
21
22
26
Normalized Na (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE


Lake Marg − SRP
21
22
26
32
33
Normalized SRP (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

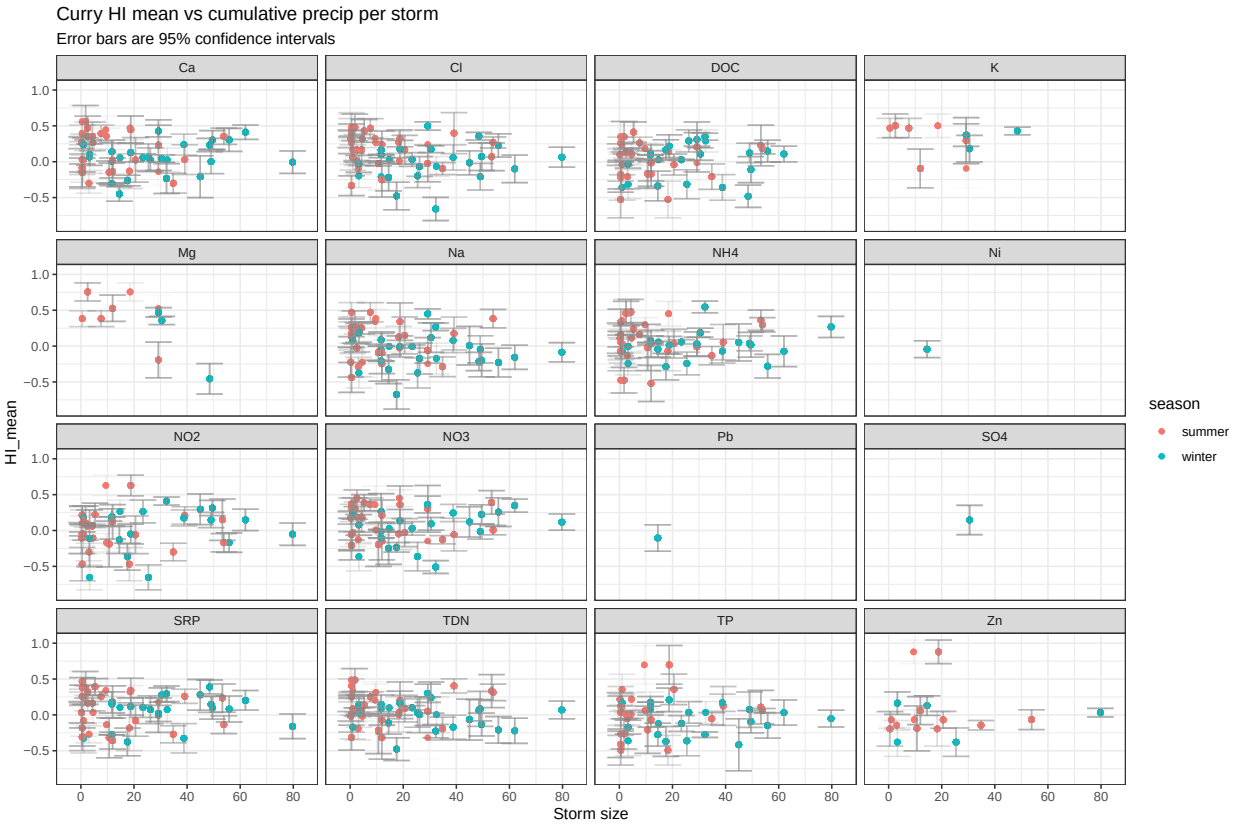

Lake Marg − TDN
21
22
26
32
33
Normalized TDN (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

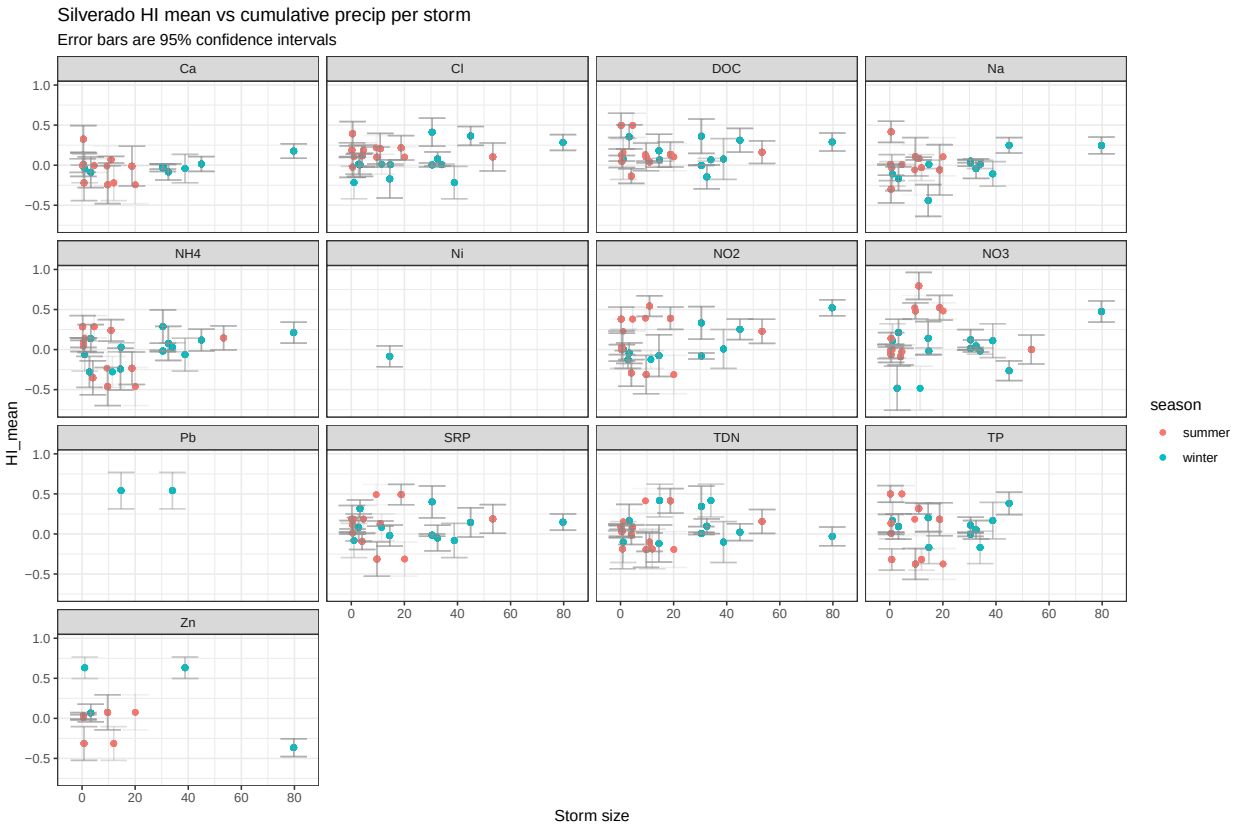

Lake Marg − TP
22
26
32
33
Normalized TP (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE
TRUE

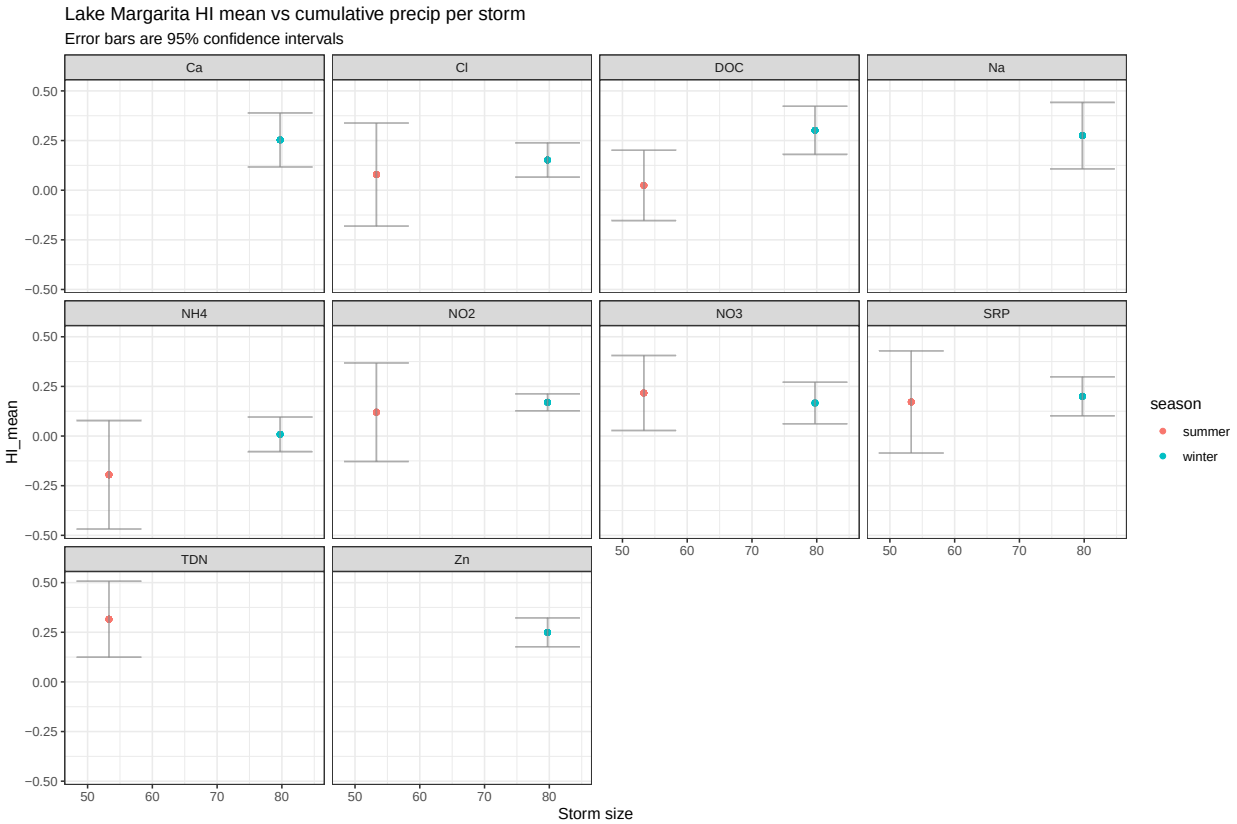

Lake Marg − Zn
21
22
26
Normalized Zn (0–1)
Normalized Stage (0–1)
Time (within storm)
outlier_analyte
FALSE

Precipitation

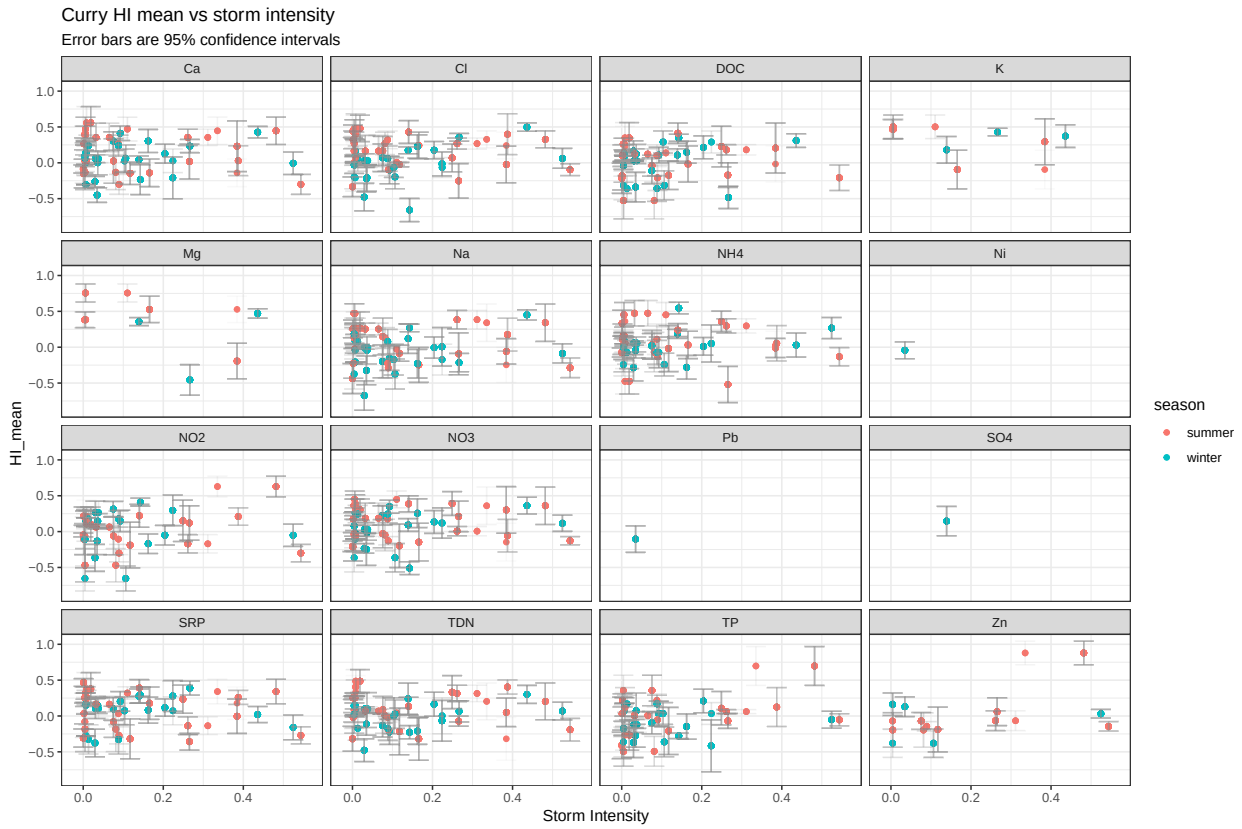
Storm size = cumulative precipitation (mm) per storm

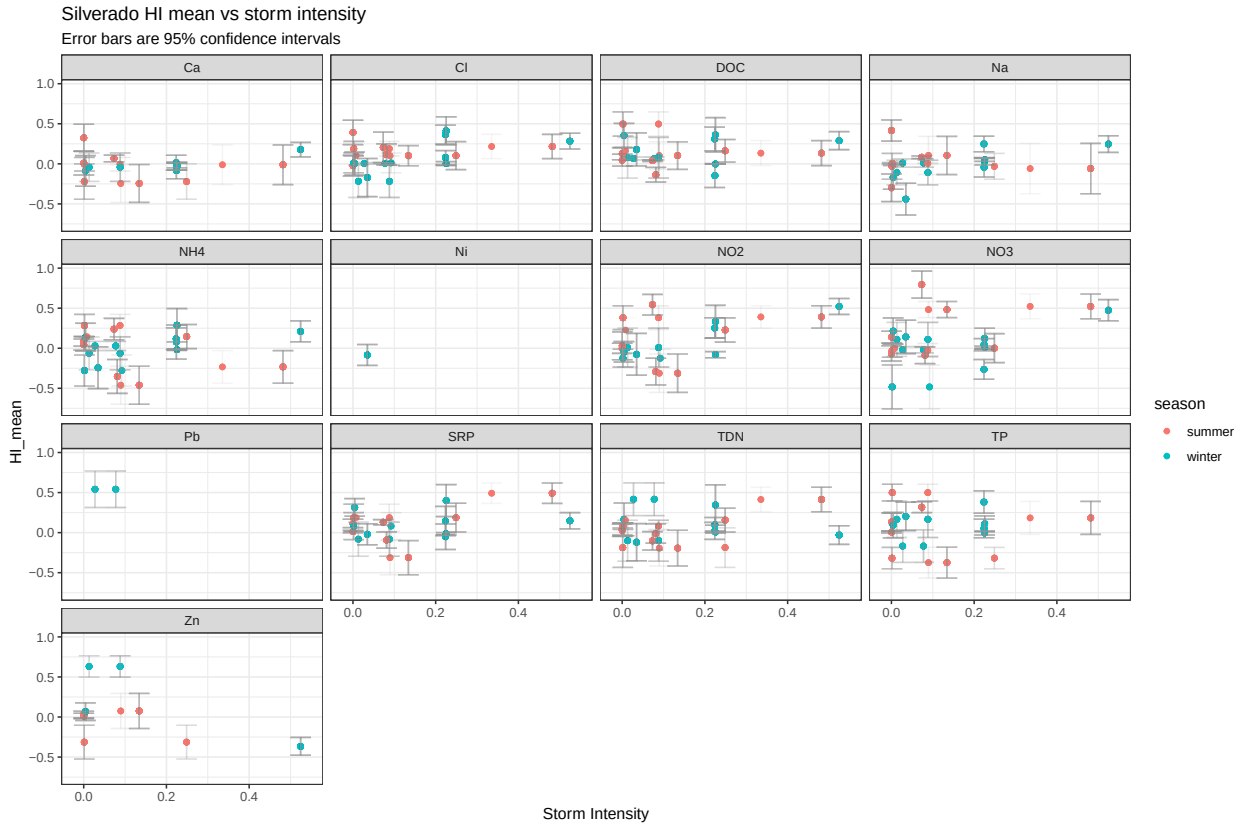


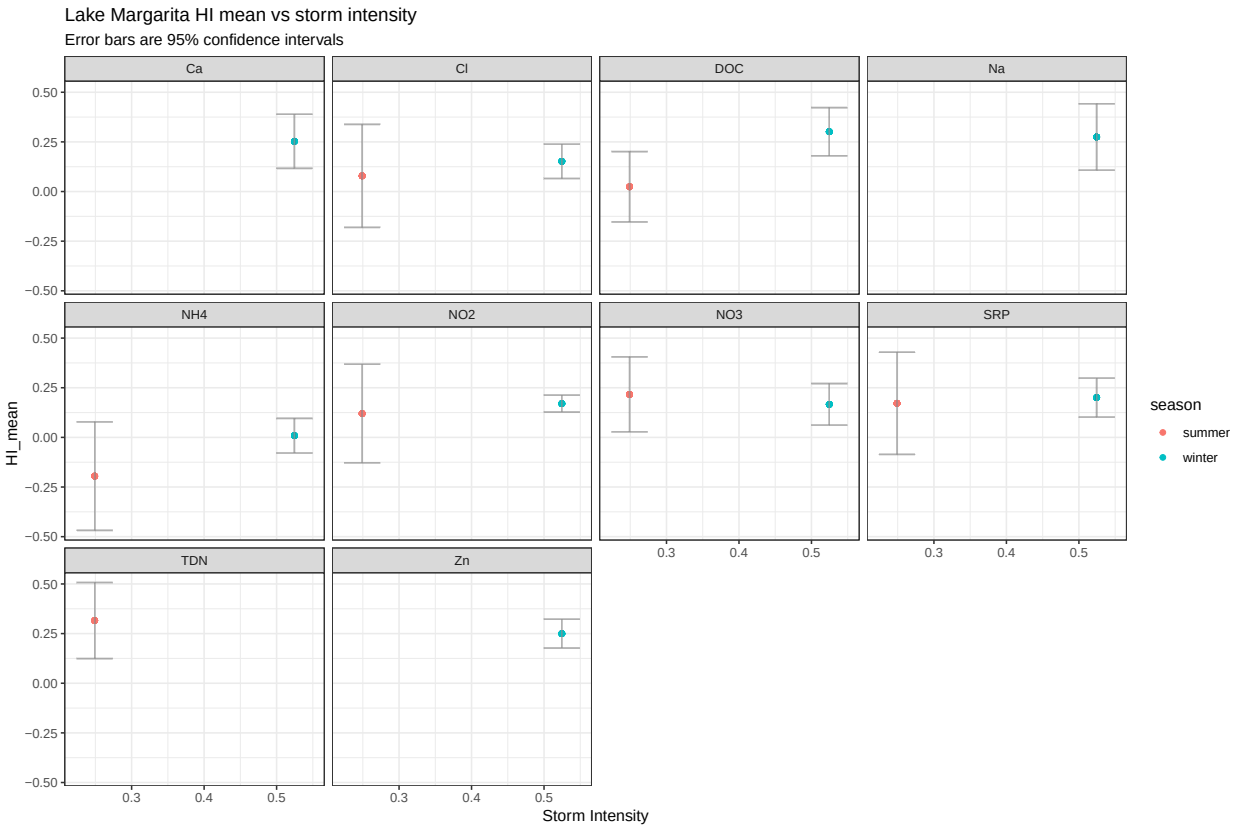




Storm intensity = cumulative precipitation/storm duration


Curry HI mean vs storm intensity
Error bars are 95% confidence intervals
Ca
Cl
DOC
K
Mg
Na
NH4
Ni
NO2
NO3
Pb
SO4
SRP
TDN
TP
Zn
HI_mean
Storm Intensity
season
summer
winter


Silverado HI mean vs storm intensity
Error bars are 95% confidence intervals
Ca
Cl
DOC
Na
NH4
Ni
NO2
NO3
Pb
SRP
TDN
TP
Zn
HI_mean
Storm Intensity
season
summer
winter


Lake Margarita HI mean vs storm intensity
Error bars are 95% confidence intervals
Ca
Cl
DOC
Na
NH4
NO2
NO3
SRP
TDN
Zn
HI_mean
0.50
0.25
0.00
−0.25
−0.50
0.3
0.4
0.5
Storm Intensity
season
summer
winter